

Japan IT Hardware & Electronics

Panasonic Deep Dive: History repeats itself... BBU from the lens of competition and BOM analysis



Kengo Makita
+81 3 6777 6977
kengo.makita@bernsteinsg.com



Ting Ming Neo
+852 2123 2554
tingming.neo@bernsteinsg.com



Do Youn Na
+81 3 6777 8519
do-youn.na@bernsteinsg.com

We initiated coverage of Panasonic with a Market-Perform rating in [Japan IT Hardware & Electronics: Japan's Physical AI Comeback - Initiating with a positive view](#). Initial investor conversations suggest this is a highly debated and differentiated call on a consensus long. We further elaborate our thoughts on how competition could play out and the technological moat of BBU.

Panasonic's Battery Backup Unit (BBU) business is its most important near-term AI data-center growth driver. BBUs provide rack-level backup power and increasingly support peak shaving as AI-server loads become denser and more volatile. Demand comes from three sequential pools: **1) new construction, 2) installed-base penetration, and 3) replacement beginning around 2030E.** We estimate the global BBU market grows from USD3.7bn today to USD16.2bn by 2032E, a 35% CAGR. Panasonic's DC Battery (BBU + CBU) revenue is expected to reach JPY1tn by 2030E.

Panasonic should retain near-term leadership, but we likely see its BBU share to decline from over 80% today to 40-50% over time. Hyperscaler qualifications, end-to-end manufacturing, automotive-grade safety expertise, and capacity expansion create customer stickiness, but not a durable moat. Competition across cells, modules, and rack integration, combined with CSP multi-sourcing and growth among neocloud and colocation providers, should broaden the qualified supplier base. We model a scenario where Panasonic's market share declines from 80% to 40% by 2032E, potentially limiting BBU revenue growth to 15-20% versus 35% for the market.

Competition affects Panasonic through lower share and margin compression. Our BOM analysis estimates an extended-duration module value of USD610 and a complete 5+1 shelf value of USD4,160. In our base case, BBU adjusted operating margin declines from 20% in 2026 to 9% in 2032E. In our bear case, margin falls to 10% in 2029E, 3% in 2030E, and negative 3% in 2031E.

Execution constraints and established competition should limit upside beyond visible BBU orders. Converting EV-battery lines into efficient BBU production could take one to two years, while established suppliers are already targeting supercapacitor-based CBUs and 800V DC architectures. Panasonic should participate, but is unlikely to capture these opportunities disproportionately, in our view.

Are we too pessimistic on valuation? We believe the current valuation already reflects a substantially stronger earnings outcome than our base case. Data-center businesses generate margins above 20%, but represent only 6% of revenue. Our two-year blended-forward EBITDA growth estimate is 14.5%, versus the more than 35% implied by Panasonic's historical growth-versus-multiple relationship to justify 9-10x EV/EBITDA. This requires broad-based growth and margin expansion, not BBU alone.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	6 Oct 2026	Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
			Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
6752.JP (Panasonic)	M	JPY	4,769.00	4,500.00	145.8%	JPY	81.17	204.60	251.87	58.8	23.3	18.9
JPL			2,709.59									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

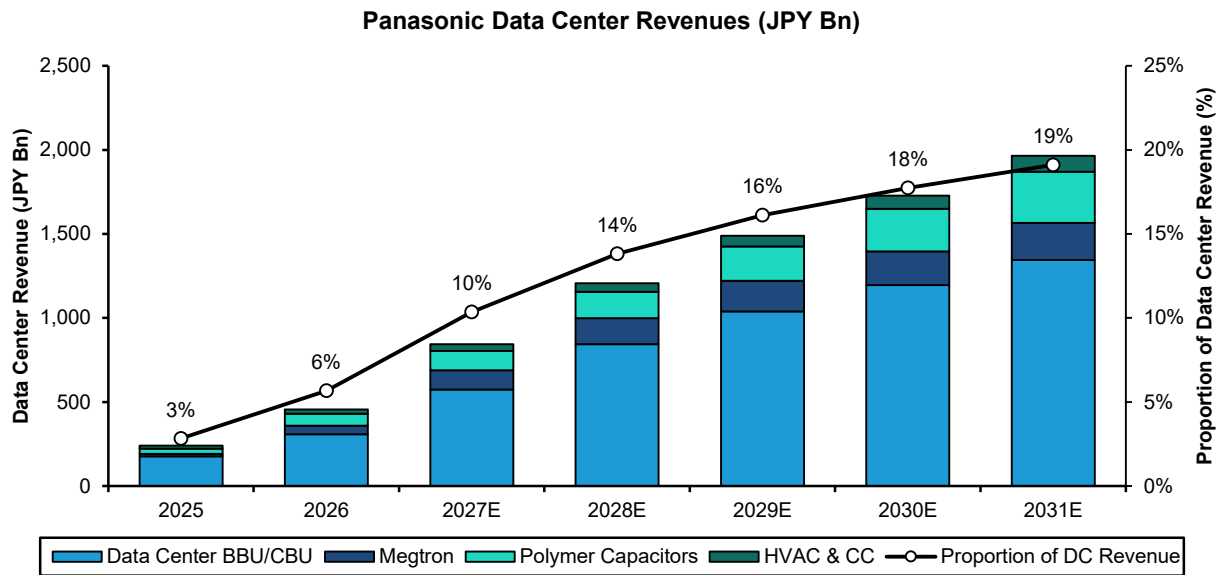
Panasonic is the most debated stock in our sector. Its high-margin data-center exposure provides meaningful near-term earnings upside, but the current 10.8x EV/EBITDA multiple reflects heightened expectations and becomes demanding when viewed against the company's expected growth and historical growth-versus-multiple relationship. **We recognize the near-term opportunity, but medium-term expectations and longer-term share erosion support our Market-Perform rating and JPY4,500 target price.**

DETAILS

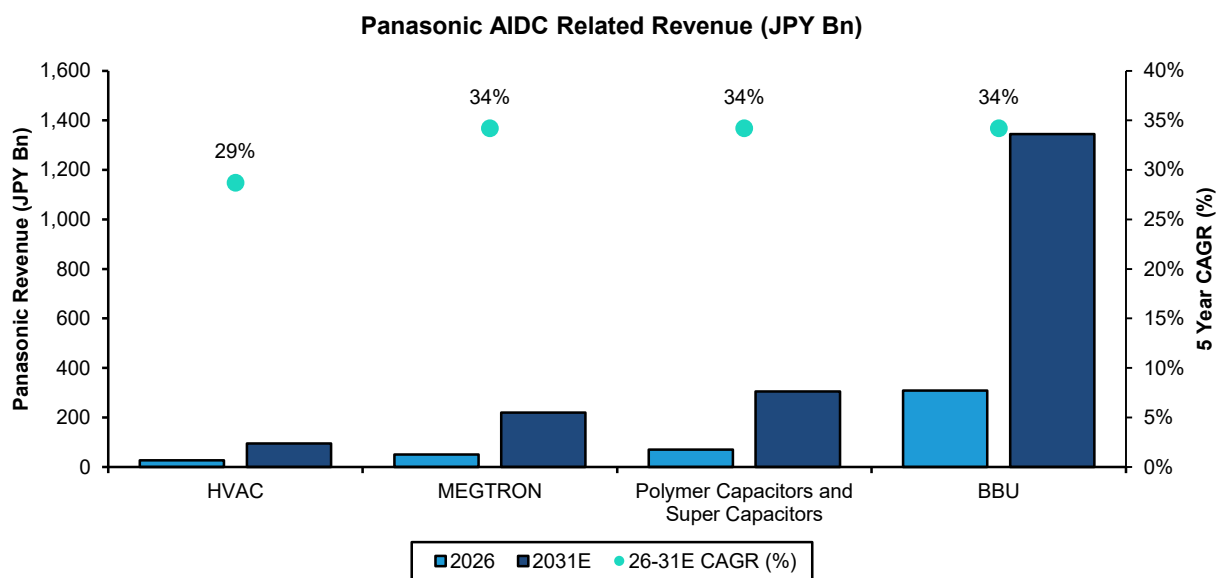
HOW LARGE IS THE BBU OPPORTUNITY?

The BBU opportunity is large, with demand supported sequentially by new AI data-center construction, penetration of the installed base, and replacement beginning around 2030E. We estimate the global BBU market grows from USD3.7bn today to USD12.7bn by 2032E, a 30% CAGR. **Panasonic's DC Battery (BBU + CBU) revenue is expected to reach JPY1tn by 2030E under the stated trajectory, making it the company's most important near-term AI data-center growth driver.**

- Panasonic's BBU business is the largest contributor to its rapidly expanding data-center portfolio.** Panasonic's broader data-center portfolio, including BBU/CBU, Megtron, polymer capacitors, and HVAC and cooling, is expected to grow at a 35% CAGR over the next five years, increasing from 6% of consolidated revenue today to nearly 20% by 2031E. BBU is the largest contributor, with revenue rising from JPY300bn today to JPY1tn by 2030E and JPY1.3tn by 2031E. Megtron and polymer capacitors each currently account for less than 1% of consolidated revenue. Because Panasonic's data-center businesses generate adjusted operating margins of 15-20%, their earnings contribution is greater than their revenue contribution, accounting for 24% of adjusted operating profit today and supporting consolidated adjusted operating-profit growth of up to 20% annually (Exhibit 1-Exhibit 5).
- BBUs are rack-level energy-storage systems designed to provide around five minutes of short-duration backup power, with systems typically sized for a 25-30-minute design margin, and increasingly support peak shaving in AI data centers.** Installed close to the server rack, BBUs deliver stable, high-power DC output during utility disturbances or transfer events, bridging the gap until UPS or other backup-power systems can fully respond. Given the high power density of AI server racks, BBUs typically use higher-output, higher-energy-density nickel-based batteries, including NCM, NMC, and NCA, despite their higher cost. BBU technology itself is not new, and Panasonic has historically supplied more than 90% of BBUs used by the five major hyperscalers. However, as AI servers become more power-intensive and their power consumption becomes increasingly volatile, both the performance requirements and strategic importance of BBUs are increasing, with peak shaving becoming an important function alongside traditional backup power. Panasonic's technology roadmap is therefore evolving toward full-rack power solutions that combine BBUs with supercapacitors: BBUs provide backup power and manage load fluctuations over several minutes, while supercapacitors respond to shorter-duration fluctuations occurring within milliseconds or seconds (Exhibit 7- Exhibit 12).
- We identify three BBU demand pools: new AI data-center construction, penetration of the installed base, and replacement.** New construction should drive demand over the next two to five years, while replacement becomes the larger and more sustainable driver after 2030E. Across these three pools, we estimate the combined BBU market grows from USD3.7bn today to USD16.2bn by 2032E, a 35% CAGR (Exhibit 13-Exhibit 17).
 - New AI data-center construction is the principal near-term driver.** We assume BBU adoption in new data centers rises from 70% today to 100% by 2027E as power stability and peak-shaving capabilities become increasingly important. Combined with 20% annual growth in data-center power demand, this increases incremental BBU capacity demand from 9GWh to 34GWh by 2032E. At an assumed module value of USD250 per kWh, we estimate the new-build market grows from USD2.5bn today to USD8.8bn by 2032E.
 - Installed-base penetration is a smaller opportunity.** We estimate that 50% of the relevant installed data-center base currently uses BBUs, rising gradually toward full penetration by 2031E. Because legacy and non-AI racks generally require less BBU capacity and limited peak-shaving functionality, we estimate the installed-base market remains at USD500-600mn through 2031E.
 - Replacement should become the largest longer-term demand pool.** With an estimated useful life of five to seven years, BBUs installed during the buildout beginning in 2025 should start generating meaningful replacement demand around 2030E. We estimate the replacement market grows from USD143mn in 2025E to USD7.4bn by 2032E, a 76% CAGR from a small base. Replacement should therefore become the largest and more sustainable demand driver as the initial data-center buildout matures.

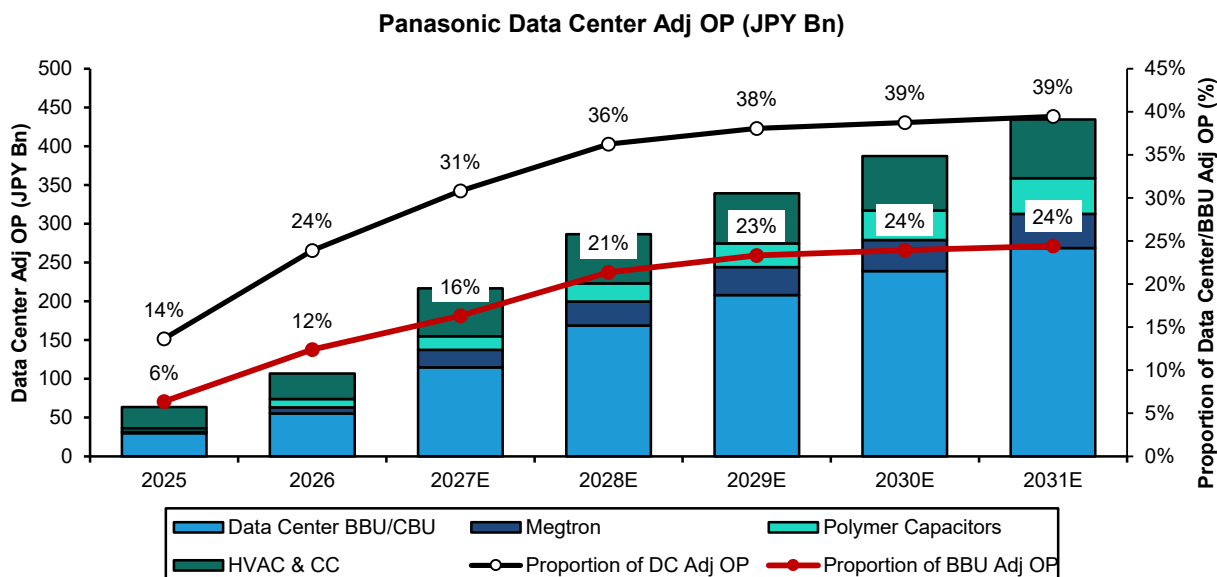
EXHIBIT 1: **Panasonic's AI data center growth inflection: +35% CAGR for key AIDC components**

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 2: **Panasonic's AI Data Center businesses are expected to grow ~30-35% CAGR in the next 5 years**

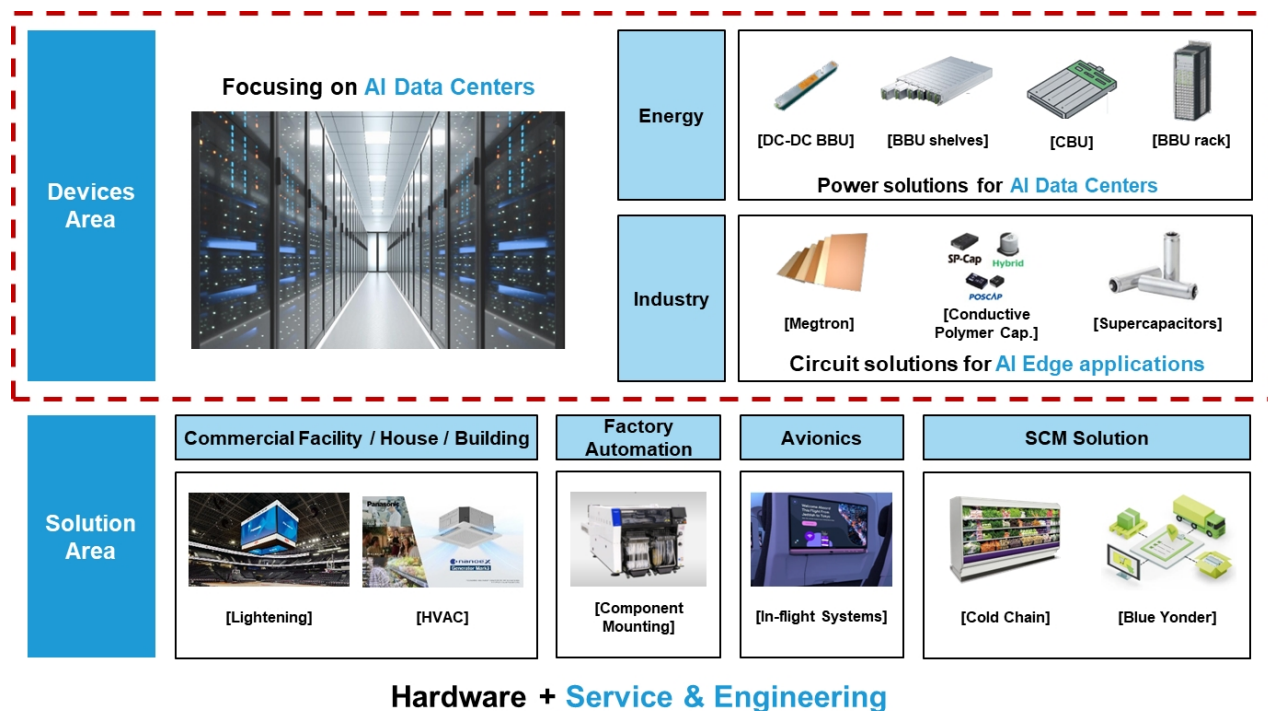
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 3: **Panasonic's DC Adj OP contribution is expected to accelerate in the short run and moderate to ~38% in 2029E**



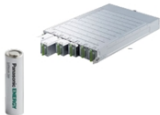
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 4: **Panasonic is focusing on its AI Data Center business spanning across energy and circuit solutions while maintaining growth in other areas**

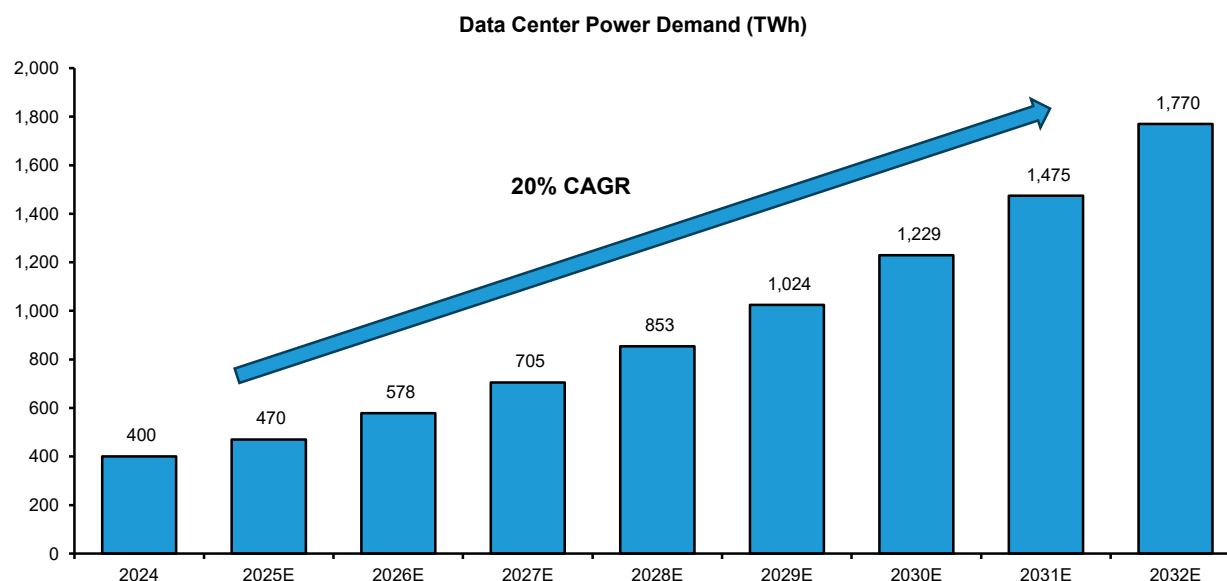


Source: Company reports, Bernstein analysis

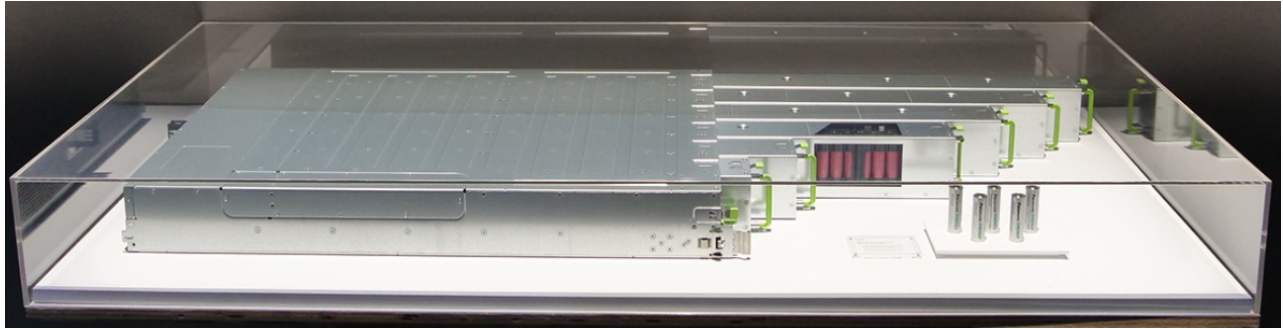
EXHIBIT 5: **Summary of Panasonic's Data Center Business accounting for ~18% of total revenue by 2031E**

	What do they do?	Market Size (USD Bn) / Market Share (%)	26-31E Revenue CAGR (%) Adj OP Margin (%)	
BBU	- Built in DC server rack energy storage system - NCM/NCA batteries, different from UPS - Improve DC efficiency through peak shaving and load absorption	- USD10Bn by 2030E 80% total market share	- Revenue CAGR: ~30% - Adj OPM: ~20%	
MEGTRON	- High-performance multilayer circuit board materials - Prevent high speed signal loss and reduce power consumption	- USD5Bn by 2030E ~6% total market share	- Revenue CAGR: 20% - Adj OPM: 15%	
Capacitors	- Regulate voltage, supplement MLCCs in circuit boards - Higher capacitance but larger compared to MLCC	- USD15Bn by 2030E ~5% estimated share	- Revenue CAGR: ~30% - Adj OPM: 15%	
HVAC	Air conditioning with data center related exposure	USD3Bn by 2030E	Revenue CAGR: 22%	

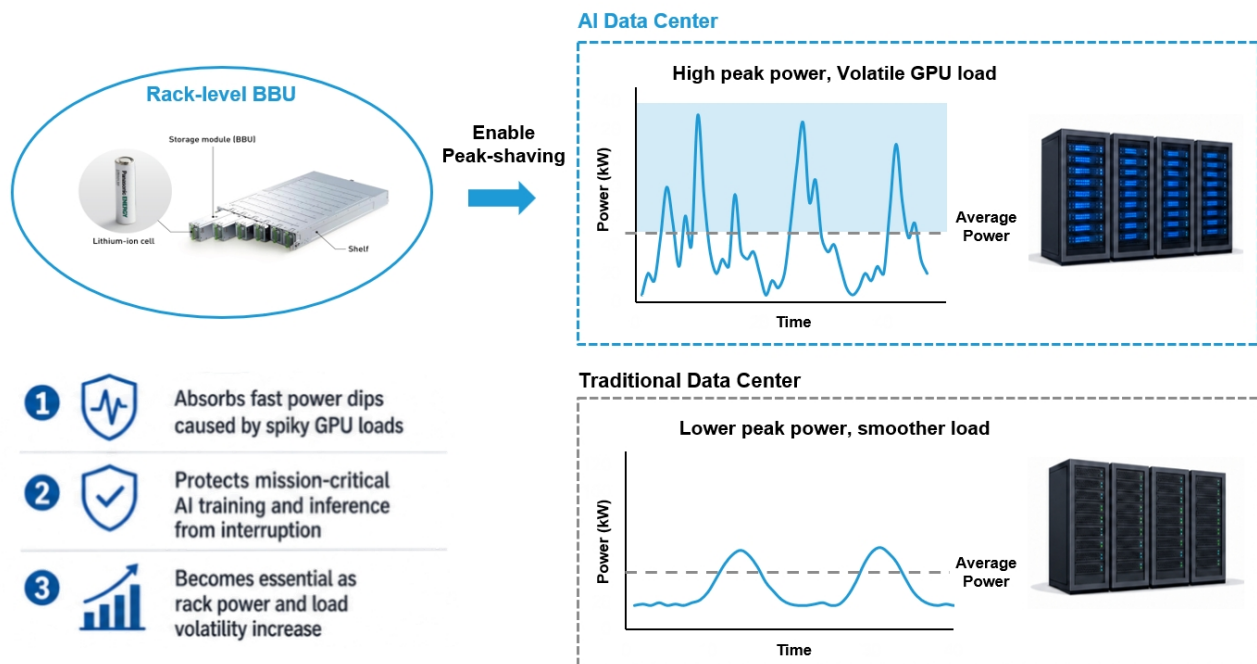
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 6: **Data center power demand is expected to grow at ~20% CAGR, driven by AI servers that consume more power**

Source: IAE, Bernstein estimates and analysis

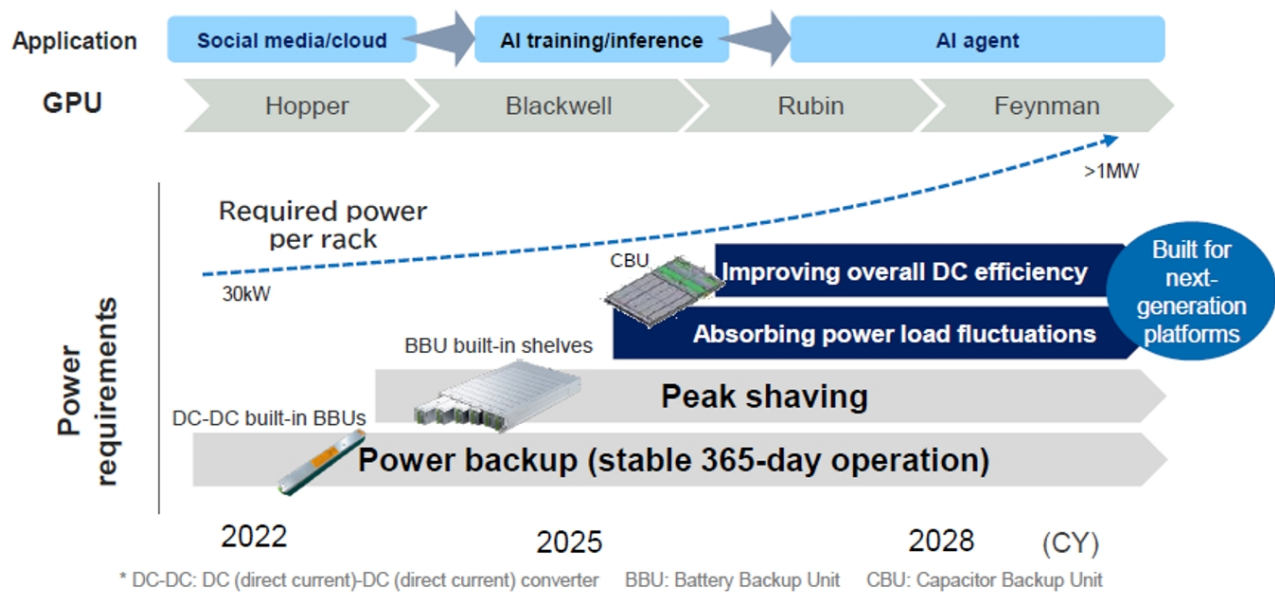
EXHIBIT 7: **BBU modules are placed in a rack in a 5+1 configuration**

Source: Panasonic

EXHIBIT 8: **Rack-level Battery Backup Units (BBU) are now essential to turn volatile AI Loads into reliable data center operations**

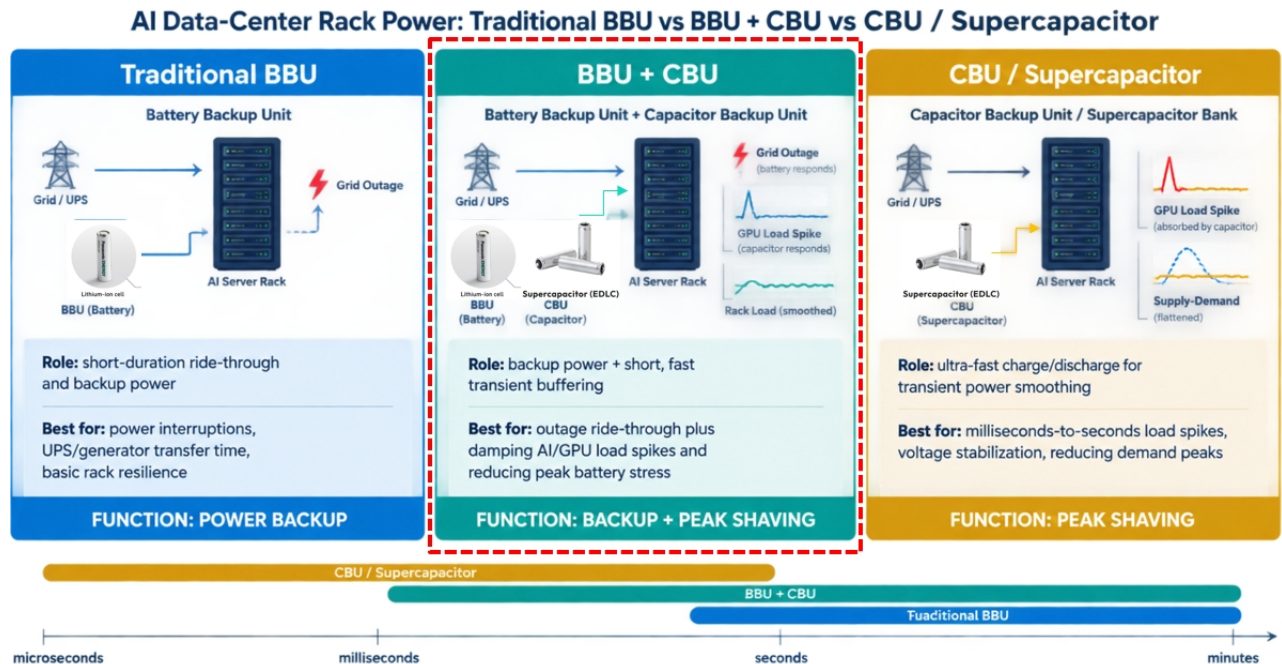
Source: Company disclosures, Bernstein analysis

EXHIBIT 9: BBUs are traditionally used for power backup purposes, but BBUs today have peak shaving functionalities for AIDC, and will evolve to CBUs where they are placed in dedicated power racks instead



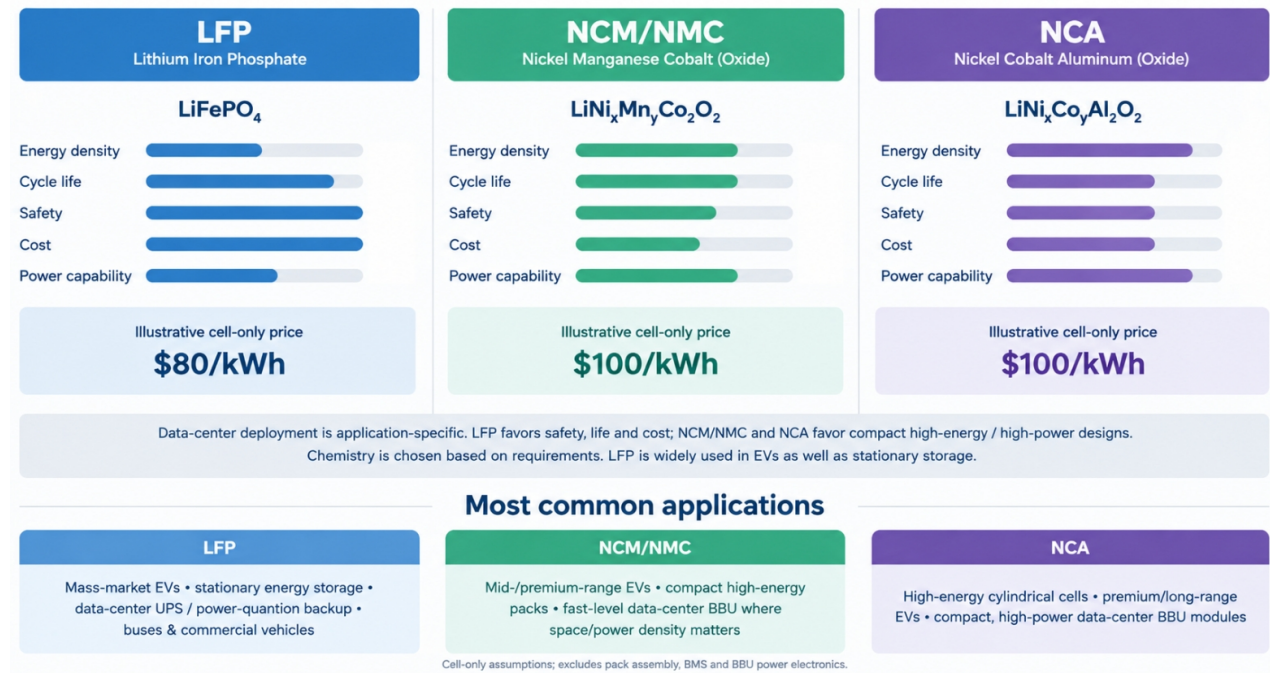
Source: Company reports, Bernstein analysis

EXHIBIT 10: Panasonic has added peak shaving functions to its traditional BBU and is working on CBUs as next generation product focusing on peak shaving



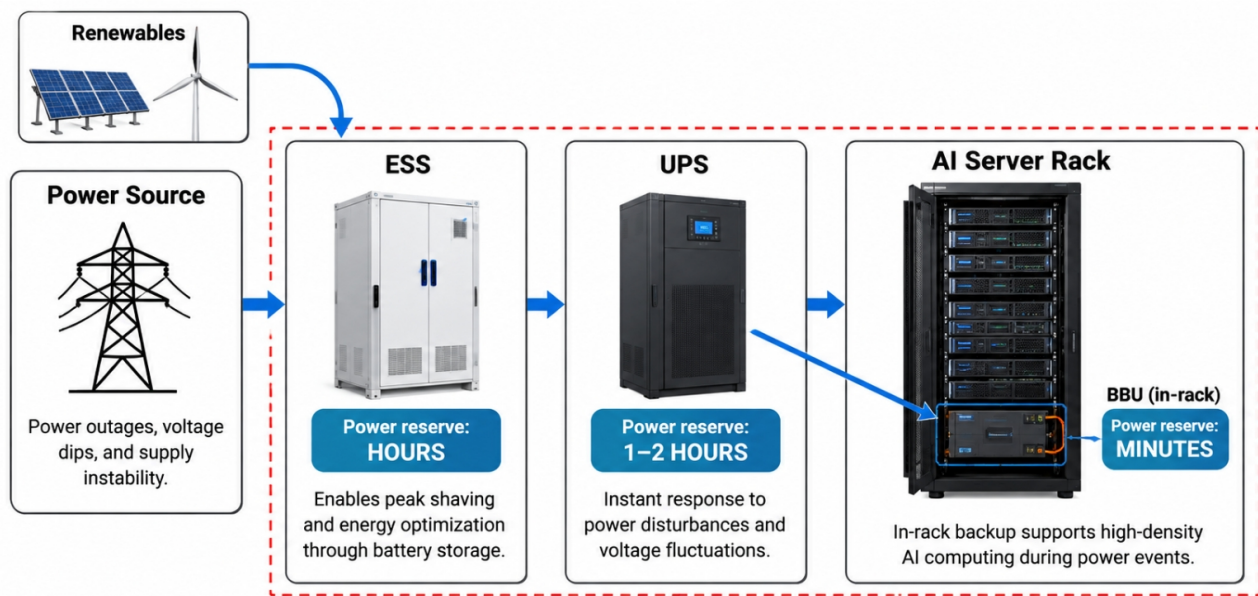
Source: Panasonic, Bernstein analysis

EXHIBIT 11: NCA and NCM are prime candidates for BBU peak shaving purposes due to its energy density but at higher costs



Source: BNEF, Bernstein analysis

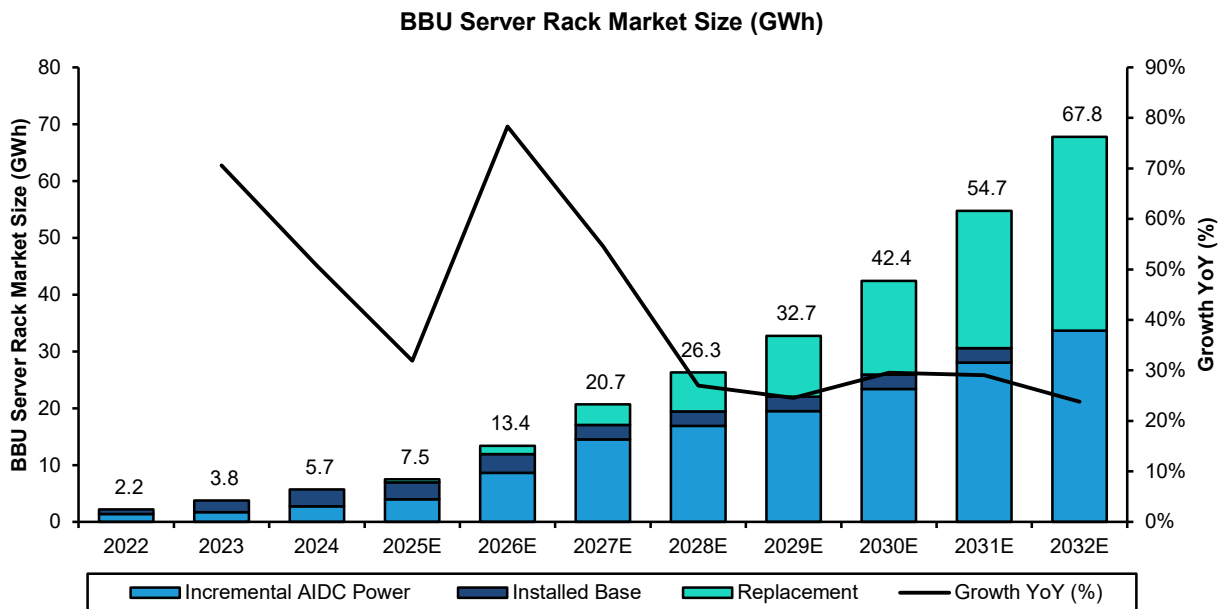
EXHIBIT 12: BBU, UPS and ESS have different roles to provide power to data centers



Rising power consumption and load volatility in AI data centers are increasing the importance of ESS, UPS and in-rack BBU.

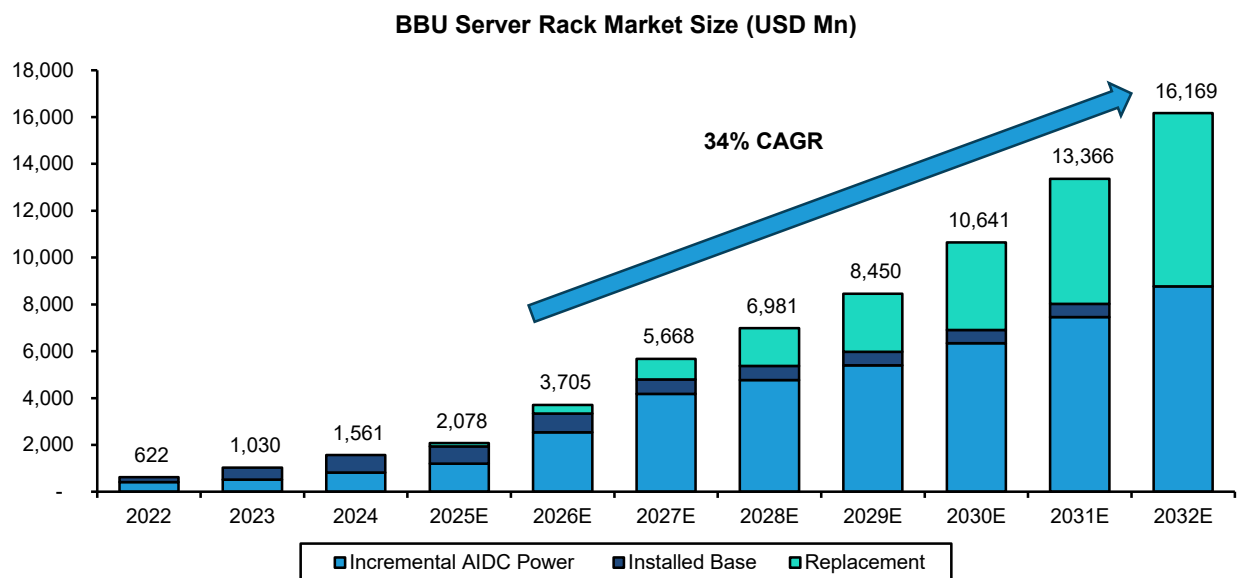
Source: Bernstein analysis

EXHIBIT 13: The BBU market is driven by 3 key drivers with incremental AIDC power the near term driver and replacement being the longer term opportunity



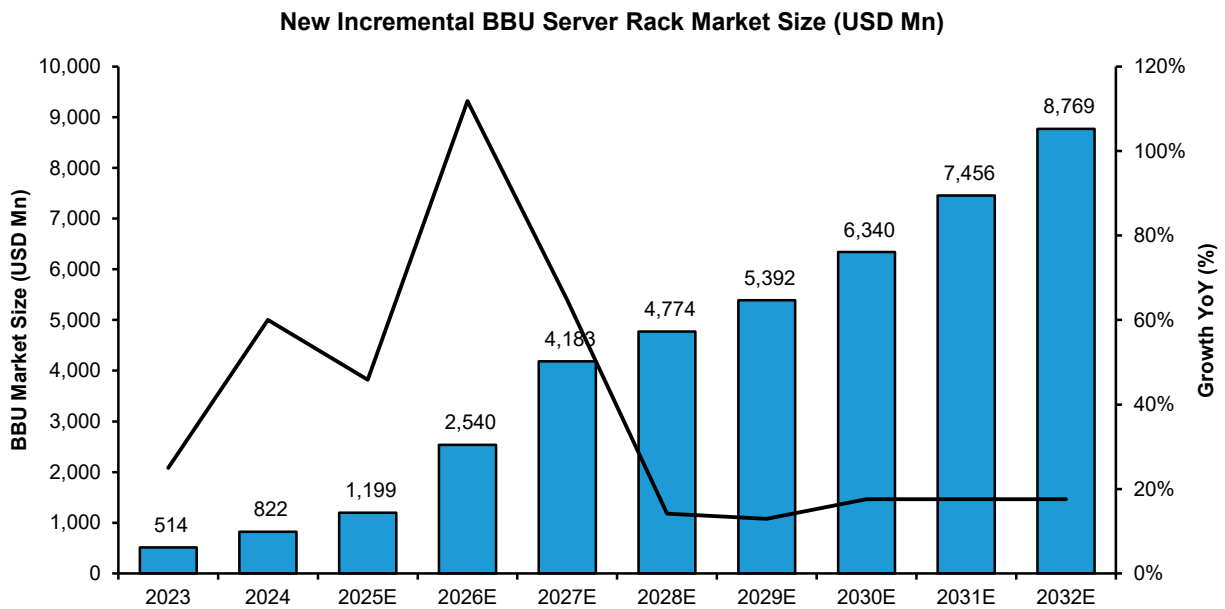
Source: IEA, Bernstein estimates and analysis

EXHIBIT 14: We expect BBU Server rack market to grow at 34% CAGR



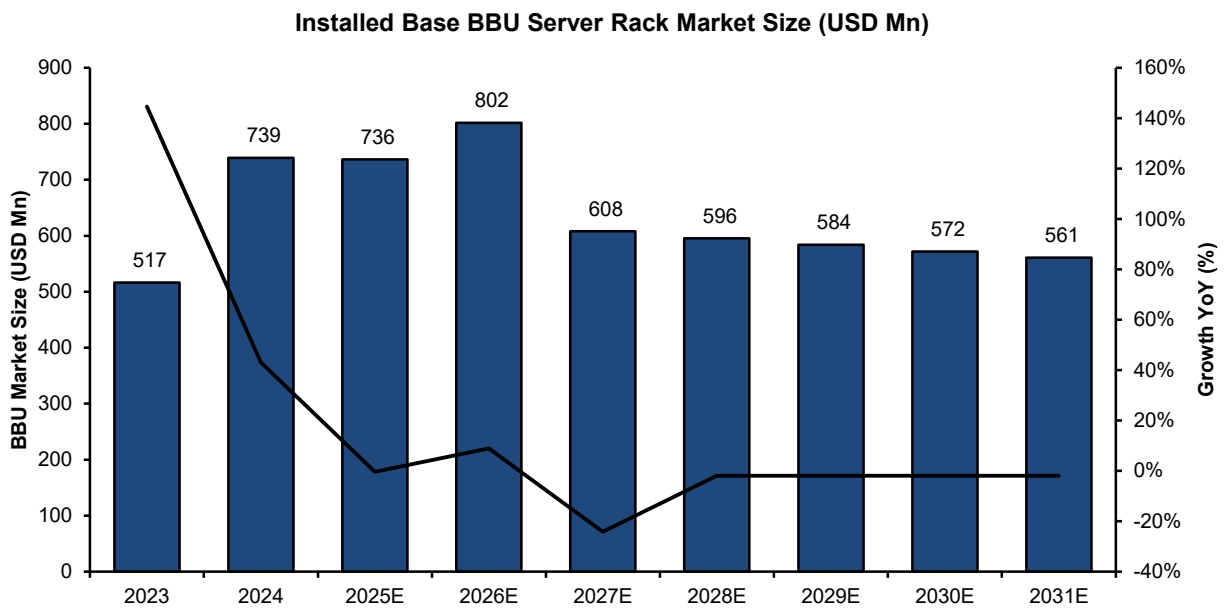
Source: IEA, Bernstein estimates and analysis

EXHIBIT 15: **New incremental AI DC server demand is the near term driver with Panasonic taking the largest share of ~80% in the near term**



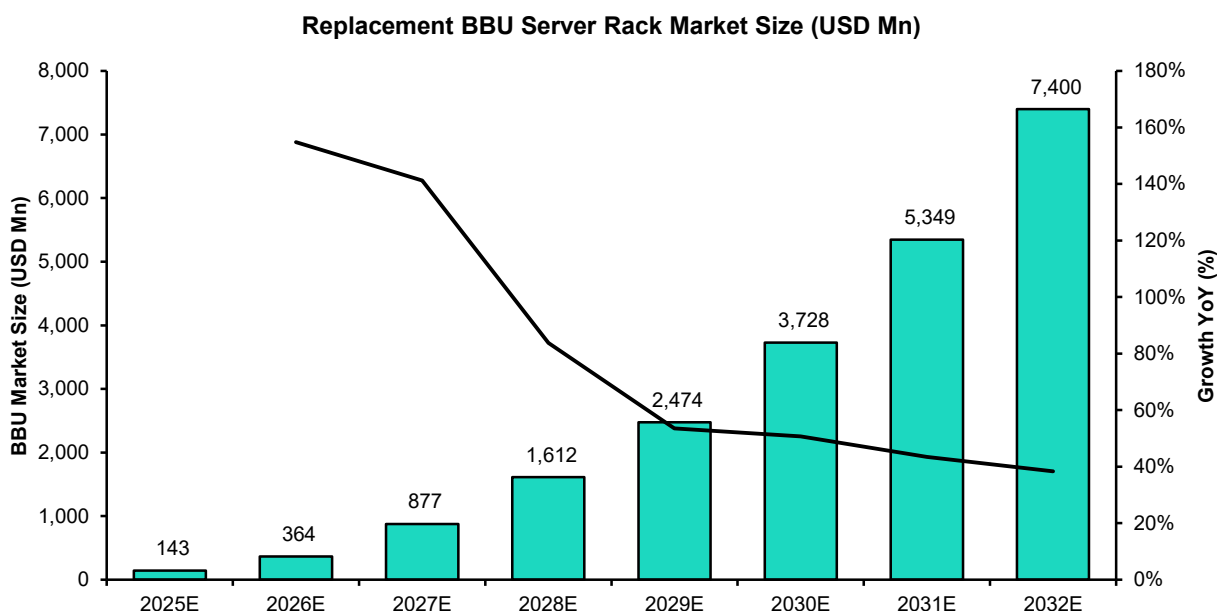
Source: IEA, Bernstein estimates and analysis

EXHIBIT 16: **Panasonic has a stronger share in installed base BBU of ~100%**



Source: IEA, Bernstein estimates and analysis

EXHIBIT 17: Replacement BBUs are a longer-term optionality driver, as older units are likely to be replaced 5-7 years after installation, although this market could eventually become a highly competitive red-ocean market



Source: IEA, Bernstein estimates and analysis

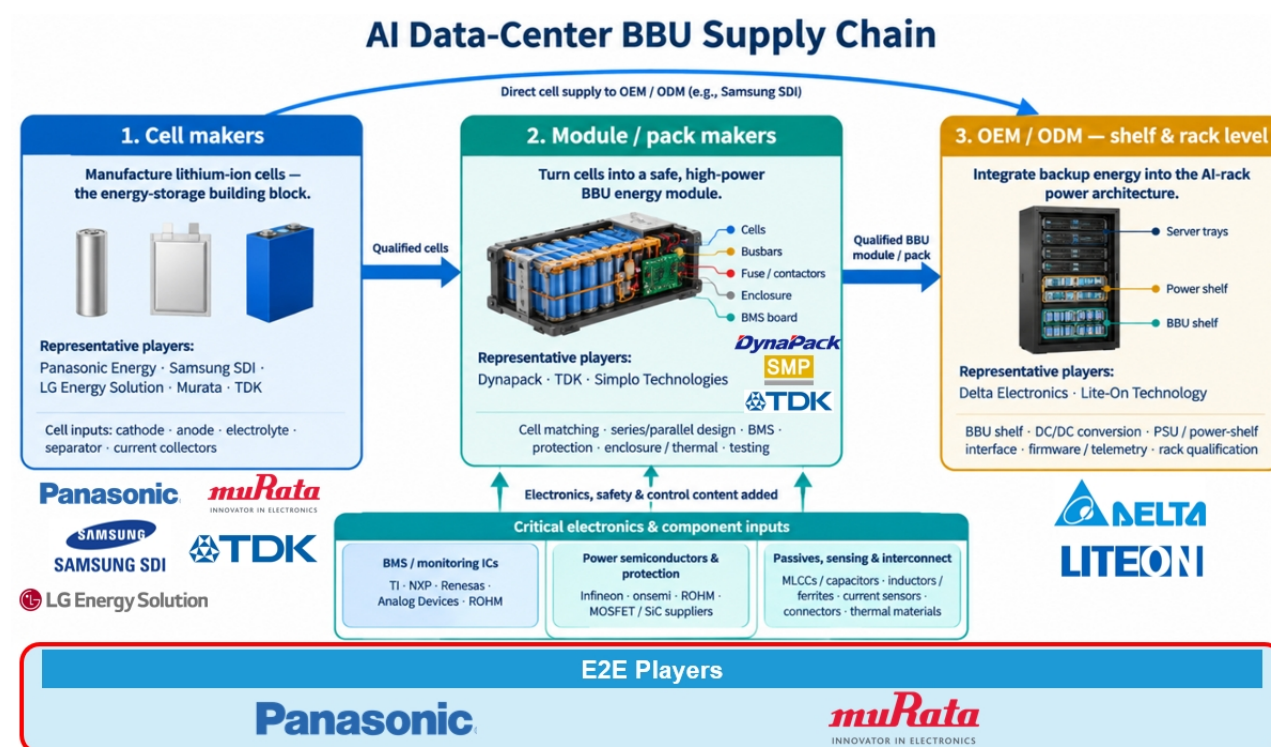
WHY IS PANASONIC THE NEAR-TERM LEADER?

Panasonic is well positioned to maintain its BBU market leadership over the near term. The company has supplied BBUs to hyperscalers since the early 2020s and currently holds an estimated market share of over 80%. We attribute this leadership to three advantages: long-standing customer qualifications, established end-to-end manufacturing and automotive-grade safety expertise, and visible capacity expansion supported by near-term demand. Together, these advantages create meaningful customer stickiness and should support Panasonic's leadership through the initial phase of the AI data-center BBU buildout.

- Panasonic has built long-standing customer relationships and a strong qualification track record with major hyperscalers.** The company has supplied BBUs to hyperscalers since the early 2020s, establishing a strong position in traditional data-center backup applications. As AI data centers generate increasingly dense and volatile power loads, these existing customer relationships and design qualifications provide Panasonic with a first-mover advantage in advanced BBUs with peak-shaving capabilities. Qualification does not create a permanent moat, but it should support meaningful customer stickiness and protect Panasonic's leadership over the near term.
- Panasonic's differentiation lies in its established end-to-end BBU manufacturing capabilities across all three stages of the supply chain.** First, **cell manufacturers** produce the battery cells that determine the chemistry, power characteristics, specifications, and safety performance of the system. Second, **module and pack manufacturers** integrate these cells with the battery-management system, power semiconductors, passive components, sensors, and interconnects to create a complete BBU module. Third, **shelf and rack assemblers** combine the BBU modules, and potentially CBU modules, into customer-specific shelf configurations before integrating them into the server rack (Exhibit 18).
- Panasonic participates across these stages, combining nickel-based battery-cell expertise, BMS and power-electronics integration, component sourcing, and module assembly within an established production model.** Its automotive-grade safety track record is particularly important for BBUs, given the high energy density and stringent reliability requirements of data-center applications. By manufacturing complete BBU modules under one roof, Panasonic can support faster production, tighter quality control, and greater system optimization, providing an early advantage over less-integrated competitors such as Delta.
- Panasonic's capacity investments provide further support for its near-term BBU leadership.** The company plans to convert portions of its EV-battery production lines in Osaka and Kansas to produce cells for data-center BBUs, while

expanding module-production capacity in Mexico. These investments indicate meaningful visibility on customer demand and should improve Panasonic's ability to serve committed orders during the initial BBU ramp. However, physical capacity and efficient production are not the same. Converting EV-battery lines requires time, and production efficiency may initially remain below mature levels. Panasonic therefore has meaningful medium-term manufacturing optionality, although conversion timing and production efficiency remain execution constraints.

EXHIBIT 18: The BBU supply chain has new legacy players entering the scene; Panasonic's edge lies in having the E2E supply chain under one roof



Samsung SDI and LG Solutions are covered by Neil Beveridge. Delta is covered by Alex Wang. Panasonic and Murata are covered by Kengo Makita. TDK and Liteon are not covered by Bernstein

Source: Bernstein analysis

EXHIBIT 19: Panasonic is investing heavily to convert battery lines for Electric Vehicles to Data Center and further expand capacity

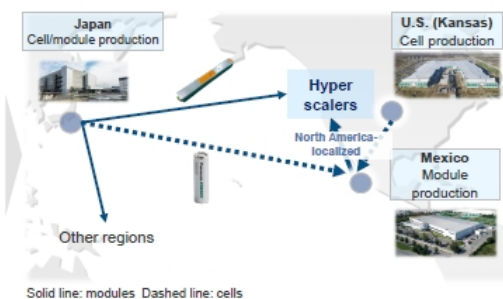
Region	Factory	Capacity (expand plan)	Production year	Application	Type	Detail
North America	Nevada	41GWh	Mass production 2017	In-vehicle	NCA	First Tesla Giga-factory
	Kansas	32GWh	Start production 2025	In-vehicle DC	Nickel based	- Initially intended for Tesla EV - Converted part of EV line toward DC application - Start mass production for DC in 2026
Japan	Tokushima	Undisclosed	Mass production 2023	DC	NCM	- Installed 21-size cell lines for industrial applications from 2023 - Key production site for DC
	Osaka (Suminoe/Kaizuka)	~7GWh (+6.5GWh)	Expanding production 2010	In-vehicle DC	NCM	- Converting EV line to DC line to respond market demand - Start shipment for DC from 2026.04 - Subaru/Mazda Vehicle line(6.5GWh)
	Oizumi	(+16GWh in 2031E)	Plan to start 2030E	In-vehicle	NCM	Planned to supply Subaru's EV, backed by Japanese government
	Wakayama	(+10GWh)	Delayed in production 2026E	In-vehicle	NCA	- Facility for 4680 type battery for Tesla - Planned to start mass production from 2024 but keep delayed in mass production

Data Center Battery Manufacturing Facilities

Source: Company reports, Bernstein analysis

EXHIBIT 20: Panasonic's capacity expansion plan indicates visibility on near-term BBU demand
Implementation plan
Japan

- Cells
 - Increase LIB production capacity by approx. 3x in FY3/29 (vs. FY3/26)
 - Repurpose the In-vehicle line at the Osaka factory (LIBs and supercapacitors) Shipment of LIB began in April 2026
- Modules (BBU, CBU)
 - Further enhance production capacity (with partner companies)


North America

- Cells [Kansas]
 - Introduce production line for data center applications (scheduled to begin mass production in FY3/29)
- Modules [Mexico]
 - The second plant is scheduled to begin mass production in FY3/27
 - Decided to establish a third plant (scheduled to begin mass production in FY3/28)
- Supply chain
 - Accelerate the shift toward a North America-based supply chain through collaboration with power suppliers (to exceed 50% by FY3/29)

Source: Company disclosures, Bernstein analysis

WHY WILL THE ADVANTAGE ERODE?

Panasonic's near-term advantages are meaningful, but we do not expect its current BBU market share of over 80% to be sustainable over the longer term. Its share has already declined from nearly 100% in traditional data-center backup applications as competition emerges across the BBU supply chain. Battery manufacturers including Murata (covered by Kengo Makita) and Samsung SDI and LG Energy Solution (covered by Neil Beveridge) possess relevant cell capabilities, while Delta Electronics (covered by Alex Wang) is entering through ODM/OEM and rack-level system integration. We expect Panasonic's share to decline to 40-50% over time as additional suppliers qualify, CSPs diversify procurement, and growth among neocloud and colocation providers weakens Panasonic's incumbent customer advantage.

- Panasonic's end-to-end manufacturing model remains a near-term advantage, but competition is emerging across every stage of the BBU supply chain.** Competitors can enter as cell manufacturers, module and pack manufacturers, shelf and rack assemblers, or broader end-to-end suppliers. Established players are already expanding from adjacent battery, power-electronics, and data-center infrastructure markets into BBU products, increasing the likelihood that Panasonic faces a broader qualified supplier base over time (*Exhibit 18, Exhibit 21*).
- Cell manufacturers: Established battery-cell manufacturers have a credible path into the BBU market.** LG Energy Solution and Samsung SDI began exploring BBU cell opportunities in mid-2026 and possess relevant expertise in nickel-based batteries, automotive-grade safety, and mass production. The prolonged EV-battery slowdown also provides an incentive to deploy existing capabilities and capacity into BBUs. Both companies are initially pursuing cell opportunities but are evaluating expansion into BBU modules. Murata and TDK, the latter of which is not covered, also bring relevant cell and component capabilities. We understand that Murata has begun preliminary supply of BBU cells to Delta Electronics, which has already secured hyperscaler business. These developments provide cell manufacturers with a path from battery supply into higher-value module production as they build customer qualifications.
- Module and pack manufacturers: An expanding group of qualified module suppliers should increase competitive pressure on Panasonic.** TDK, Dynapack, and Simplo Technologies, all not covered, can combine third-party cells with BMS, power semiconductors, passive components, sensors, and interconnects to produce complete BBU modules. Their experience in PC and electronic-device batteries provides relevant battery-pack, BMS, and safety capabilities. TDK has received customer approvals and plans to begin volume production of high-voltage DC BBUs from 3Q26. Dynapack expects to begin mass production of 8kW and 12kW BBUs and shipments to a new CSP in 4Q26, followed by products above 15kW and other high-voltage DC solutions in 2027. Simplo Technologies is developing data-center-specific BBUs using Samsung SDI cells, consistent with customer requirements that exclude Chinese-made batteries. These developments indicate that module-level competition is progressing from product development to qualification and commercial production.
- Shelf and rack assemblers: Established CSP relationships and system-integration capabilities make rack-level suppliers credible competitors.** Delta Electronics and Lite-On can integrate BBU modules and other power-management components into complete shelf- and rack-level systems. Delta is the more advanced challenger, with an estimated 10-20% BBU share and an 800V DC power-rack solution already developed. Lite-On has begun deploying BBU power racks and plans to deliver next-generation systems from 2H26. As established OEM/ODM suppliers, both can leverage their CSP relationships to participate in new product generations while sourcing components from multiple vendors to optimize cost, performance, supply availability, and system architecture. These capabilities should help rack-level suppliers qualify alternative solutions and potentially gain share from Panasonic.
- End-to-end suppliers: Panasonic and Murata are end-to-end BBU participants, although Panasonic retains stronger first-mover, qualification, and scale advantages.** Panasonic's differentiation therefore lies not in exclusive end-to-end capabilities, but in its established production base, automotive-grade safety record, hyperscaler qualifications, and experience integrating cells, BMS, power electronics, components, and module assembly. Murata's cell, control-unit, and power-supply capabilities nevertheless show that Panasonic's integrated model can be replicated. Panasonic should retain an advantage during the initial market buildout, but that advantage should narrow as other suppliers develop integrated capabilities internally or through partnerships.
- BBU qualification is demanding, but the underlying technological barriers do not appear insurmountable for established competitors.** Panasonic's advantage lies in safely integrating high-density cells, BMS, power electronics, and related components within the constrained space of a server rack. These capabilities are important given the stringent safety, reliability, and power requirements of data-center applications, but they are not exclusive to Panasonic. Samsung SDI, LG Energy Solution, and Murata possess relevant expertise in battery chemistry, automotive-grade safety, and mass production. BMS and system-integration capabilities can also be developed internally or through partnerships with specialized module and power-system suppliers, although validation and customer qualification require time. Competitors should therefore be able to combine existing capabilities with supply-chain partnerships to narrow Panasonic's advantage as additional suppliers qualify.
- CSP preference for multi-vendor sourcing should further limit Panasonic's ability to sustain its dominant market share.** Panasonic's customer relationships and design qualifications create near-term stickiness, but CSPs typically diversify

procurement once credible alternatives meet functional, safety, and reliability requirements. This reduces single-supplier dependence, strengthens supply-chain resilience, and increases flexibility across pricing, capacity, and system architecture. We therefore expect CSPs to allocate BBU orders across multiple qualified suppliers rather than maintain Panasonic's current dominance. Supplier structures in DRAM, optical components, and copper-clad laminates support this broader diversification pattern, although they do not determine the eventual BBU market-share structure (*Exhibit 23*).

- **Growth outside the leading hyperscalers could further dilute Panasonic's market share.** Faster-growing neocloud and colocation providers have fewer legacy relationships and design qualifications with Panasonic, making them more open to alternative suppliers, configurations, and power architectures. Panasonic could therefore retain a strong position with existing hyperscalers while losing share in the broader market as the customer base expands.
- **We expect Panasonic's share to decline to 40-50% under a CSP multi-vendor strategy, with different share paths across the three demand pools.** In new AI data-center construction, evolving server and power architectures give CSPs greater flexibility to qualify alternative suppliers, while neocloud and colocation growth broadens the market beyond Panasonic's strongest hyperscaler relationships; we therefore expect Panasonic's new-build share to decline from 80% to 40% by 2032E. In installed-base penetration, existing qualifications make Panasonic more difficult to displace, supporting a more gradual decline from 100% to 80%, although this demand pool becomes less important as penetration approaches its ceiling. In replacement demand, we expect Panasonic's share to decline from 60% to 40% by 2032E. Replacement becomes increasingly important as the initial BBU installation base reaches the end of its five-to-seven-year useful life, but more mature specifications and a broader qualified supplier base should increase competition across price, performance, capacity, and delivery. Combined, these assumptions reduce Panasonic's total BBU share to 40% by 2032E. **Although the overall market is modeled to grow at 30% annually, share erosion limits Panasonic's BBU revenue growth to 15-20%, materially below the industry rate (*Exhibit 22-Exhibit 25*).**

EXHIBIT 21: **Panasonic is the only E2E BBU provider, but established companies across the supply chain are seeing strong demand for BBU products**

Supply Chain	Company	Time Period	Comments
E2E	Panasonic	Sep-26	3x BBU Capacity Expansion. JPY1.5Tn contract revenue by 2030E.
E2E/Cell	Murata		E2E supplier covering cell production, smart battery control units, and power supply units in a shelf. Cell supplier to OEM and Module makers too.
Cell	Samsung	Aug-26	As BBU sales increase due to expanded data center investments, cylindrical battery production is rising, and utilization rates are high
Cell	LG	Jul-26	Starts producing 2170 cylindrical batteries
Cell/Module	TDK	Jun-26	In the BBU market, we have already received the go-ahead from several customers. We plan to start volume production of high-voltage DC battery backup units from the third quarter.
Module	Dynapack	Jun-26	Demand for BBUs and order visibility for the second half remain solid, and the company expects to begin shipments to a new cloud service customer in the fourth quarter. The company expects to start mass production of 8kW and 12kW BBUs in the fourth quarter, and begin entering volume production of those with capacities above 15kW and high-voltage direct current-related products next year
Module	Simplo Technologies	Jul-26	Simplo's preference for Samsung SDI reflects not only their long-standing business relationship but also Samsung SDI's technology tailored to data center requirements and US policies aimed at excluding Chinese-made batteries
OEM	Delta	Mar-26	800V DC power rack embedded with 80kW BBU per shelf
OEM	Lite-On	Jul-26	Started deploying BBU Power racks. To deliver next generation racks from 2H2026.

E2E refers to "End to End" manufacturer, producing the cells and assembly to deliver the full BBU unit

Source: Company reports, Bernstein analysis

EXHIBIT 22: We believe Panasonic is strongest in installed base followed by new incremental AIDC built and lastly replacement where the barriers are most easily broken

BBU Share

Total BBU Module (USD Mn)

	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E
Panasonic	-	539.95	927.51	1,396.69	1,781.05	2,971.26	4,210.57	4,653.89	4,815.34	5,119.15	5,943.46	6,467.58
Delta Electronics	-	82.19	102.74	164.38	275.45	678.98	1,116.68	1,497.86	2,177.77	3,106.45	3,925.75	4,850.68
Samsung SDI/Others	-	-	-	-	21.40	54.54	340.72	829.51	1,456.53	2,415.21	3,496.88	4,850.68

Incremental AIDC Power

Panasonic	80%	80%	80%	80%	80%	80%	75%	70%	60%	50%	45%	40%
Delta Electronics	20%	20%	20%	20%	20%	20%	20%	20%	25%	30%	30%	30%
Other Potential Entrants	0%	0%	0%	0%	0%	0%	5%	10%	15%	20%	25%	30%

Installed Base

Panasonic	100%	100%	100%	100%	100%	90%	90%	85%	80%	80%	80%	80%
Delta Electronics	0%	0%	0%	0%	0%	10%	10%	10%	15%	15%	15%	15%
Other Potential Entrants	0%	0%	0%	0%	0%	0%	0%	5%	5%	5%	5%	5%

Replacement

Panasonic	60%	60%	60%	60%	60%	60%	60%	50%	45%	40%	40%	40%
Delta Electronics	25%	25%	25%	25%	25%	25%	25%	30%	30%	30%	30%	30%
Other Potential Entrants	15%	15%	15%	15%	15%	15%	15%	20%	25%	30%	30%	30%

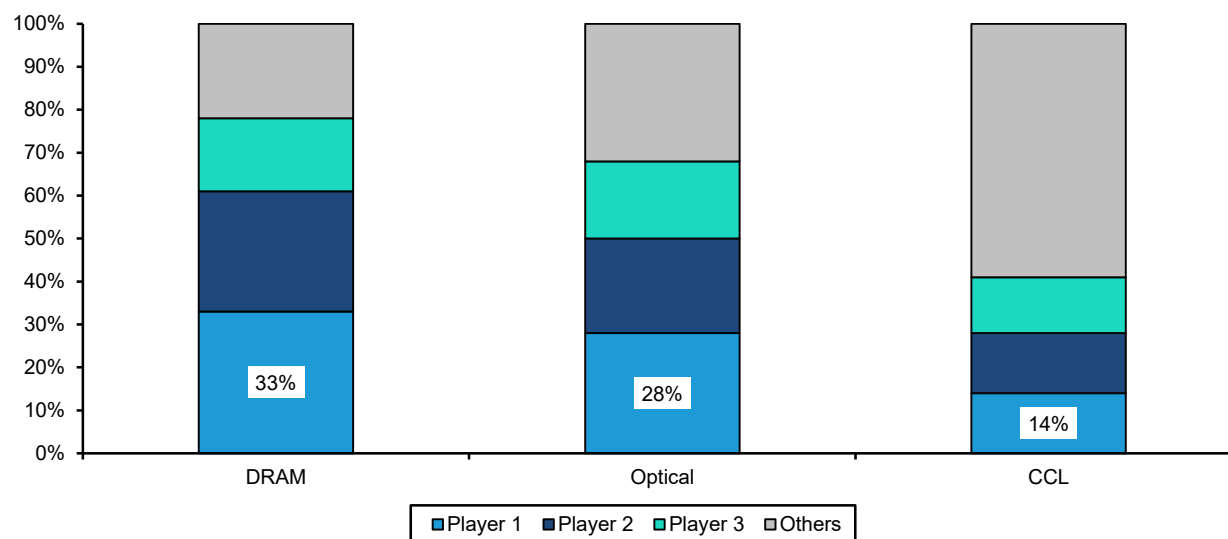
Total Market Share (%)

	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E
Panasonic		87%	90%	89%	86%	80%	74%	67%	57%	48%	44%	40%
Delta Electronics		13%	10%	11%	13%	18%	20%	21%	26%	29%	29%	30%
Other Potential Entrants		0%	0%	0%	1%	1%	6%	12%	17%	23%	26%	30%

Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 23: Hyperscalers generally prefer a multi-vendor strategy in their products to diversify risks even as customer-vendor relationships remain strong

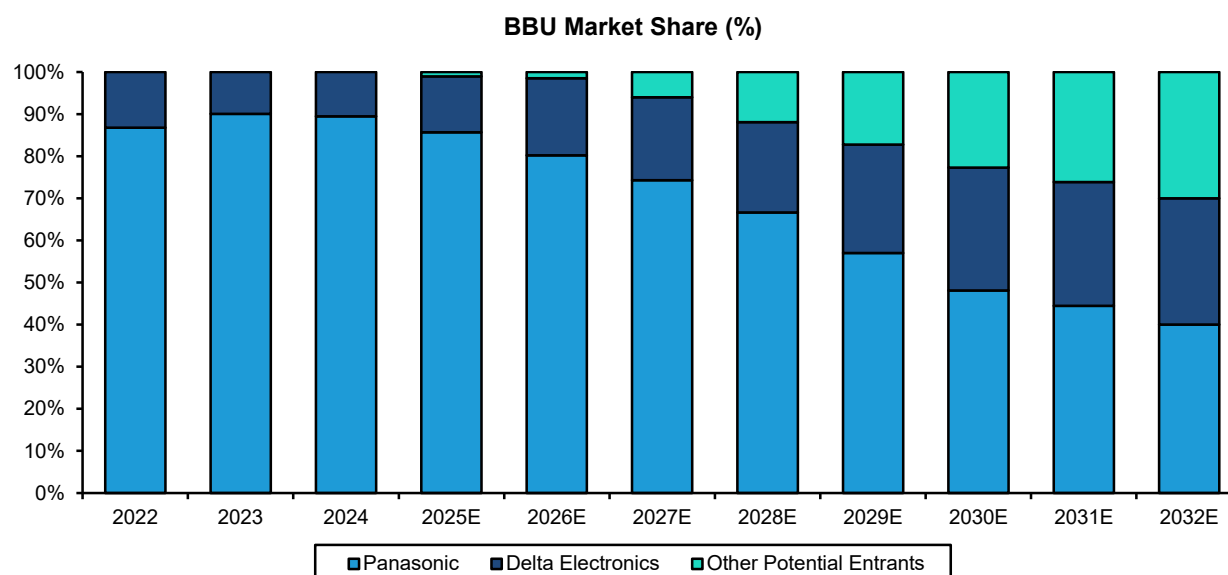
Market Share of Data Center Component Players (%)



DRAM players are Samsung, SK Hynix, Micron (Covered by Mark Li); Optical players are Zhongji Innolight, Eoptolink and Coherent (All not covered); CCL players are Kinboard, Shengyi Tech and EMC (All not covered).

Source: Company disclosures, Prismark, Bernstein Global Memory Team, Bernstein Asia Tech Hardware Team, Bernstein analysis

EXHIBIT 24: We believe Panasonic has a first mover advantage especially in near term incremental AIDC related demand, but longer term replacement and installed base penetration demand is likely to be a highly competitive area



Delta Electronics (Covered by Alex Wang)

Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 25: We estimate Panasonic's AIDC BBU business will grow around 15-20% CAGR from share loss even as the industry grows at 30% CAGR in the next 7 years

		Industry BBU Capacity 7 Year CAGR (%)				
		10%	20%	30%	40%	50%
Market Share (%)	20%	2.5%	5.0%	7.5%	10.0%	12.5%
	40%	5.0%	10.0%	15.0%	20.0%	25.0%
	60%	7.5%	15.0%	22.5%	30.0%	37.5%
	80%	10.0%	20.0%	30.0%	40.0%	50.0%
	100%	12.5%	25.0%	37.5%	50.0%	62.5%

Source: Company reports, Bernstein estimates and analysis

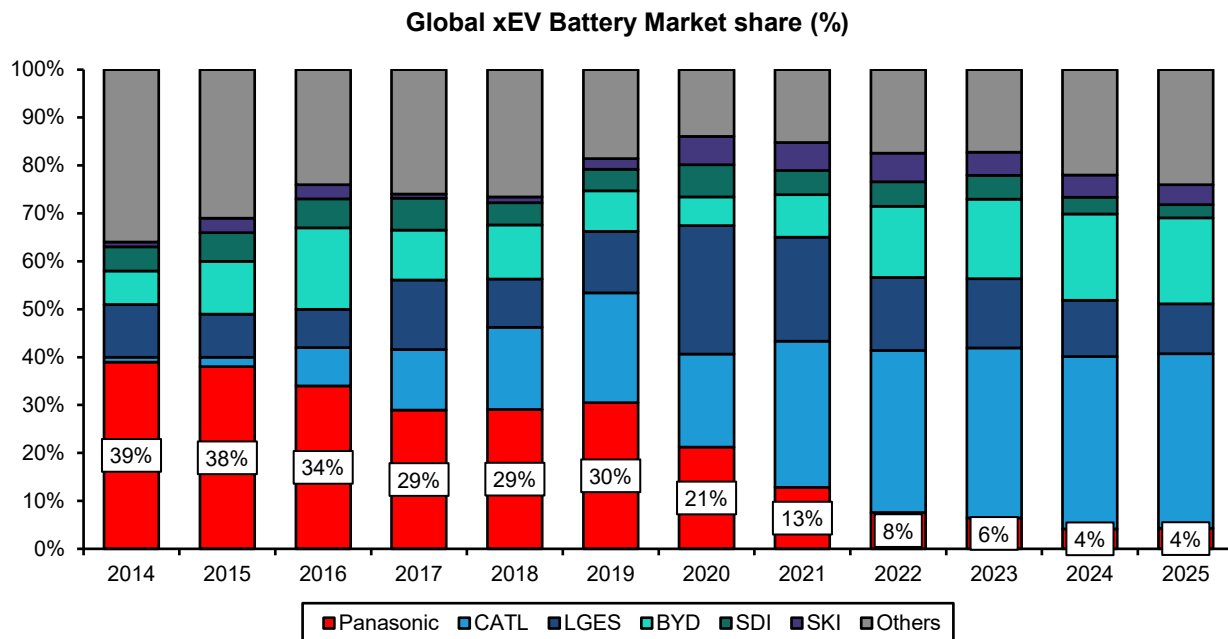
Panasonic's execution constraints and intensifying competition should limit its ability to capture upside beyond its visible BBU orders. Panasonic's EV-battery history shows that early leadership may not translate into durable market share once competition shifts from initial qualification toward pricing, capacity, and mass production. A similar risk applies to BBUs: converting EV-battery lines into efficient BBU production will take time, while adjacent opportunities in supercapacitors, CBUs, and 800V DC power architectures are already being targeted by established competitors. We therefore see limited scope for Panasonic to capture these emerging opportunities disproportionately.

- Panasonic's EV-battery history highlights the risk that early leadership may not translate into durable market share.** Panasonic's global xEV battery share declined from 30% in 2019 to 4% in 2024-2025 as the company fell behind later entrants in pricing and capacity expansion. We see a risk of this pattern repeating in the BBU market as new competitors enter and competition shifts from initial customer qualification toward pricing, production scale, and capacity expansion. Even in North America, where Chinese battery manufacturers have had a more limited presence, Panasonic has faced increasing competition from Korean battery makers. The slowdown in EV demand has also created excess capacity that Korean battery manufacturers could deploy into new applications, including BBUs. Panasonic's early BBU leadership may therefore prove temporary as the market expands and competition intensifies (*Exhibit 26, Exhibit 27*).
- Panasonic's BBU capacity expansion may take longer than expected to translate into efficient production.** Panasonic plans to convert portions of its Osaka and Kansas production lines from EV cells to data-center BBU cells, with

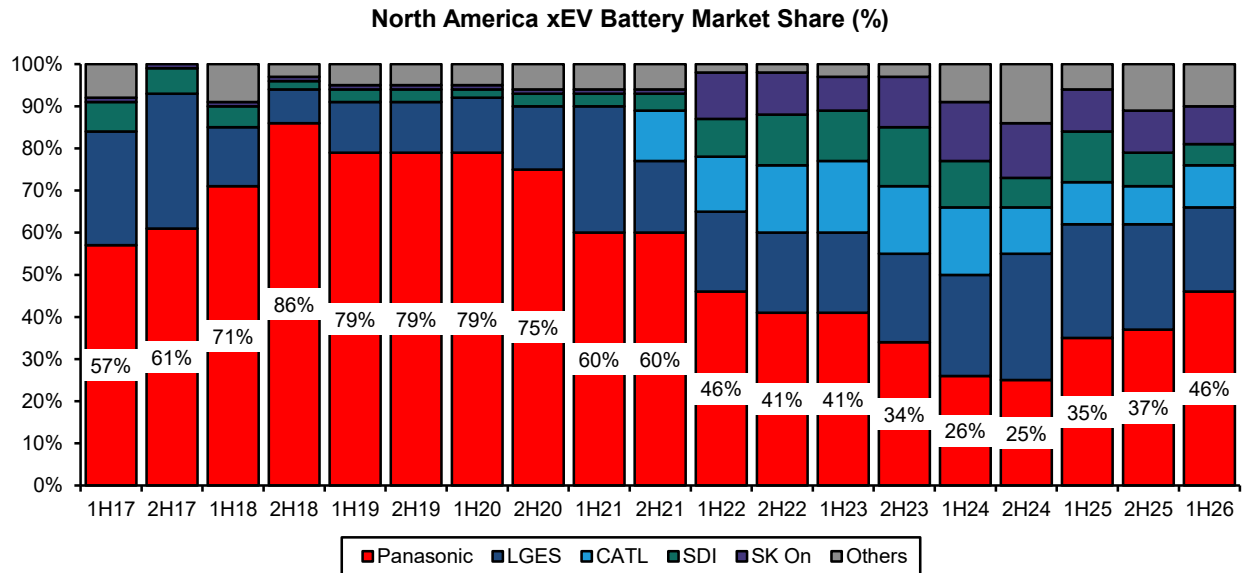
total plant capacity of 7GWh in Osaka and 32GWh in Kansas. We estimate that converting the lines will take one to two quarters, while transferring manufacturing know-how and reaching mature production efficiency could take one to two years. Although BBU cells are likely to use similar NCA or NCM materials, differences in packaging and production processes mean that EV-battery capacity will not convert into BBU capacity on a one-for-one basis. **Execution challenges during the ramp-up of the Kansas facility further illustrate the time required to translate physical capacity into efficient production.** BBU capacity could therefore remain tight over the next few years despite Panasonic's substantial physical capacity and ongoing investments.

- **Supercapacitor-based CBUs could become Panasonic's next data-center growth opportunity, although commercial adoption remains at an early stage and competition is emerging from established supercapacitor suppliers.** Panasonic is developing CBU-equipped shelves, with mass production scheduled to begin in 2027. Unlike lithium-ion BBUs, which provide backup power and support longer-duration peak shaving, CBUs use supercapacitors to rapidly absorb and release power, smoothing the short-duration load fluctuations generated by AI servers. Maxwell, VINATech, Nippon Chemi-Con, LS Materials, TDK, and Musashi Energy Solutions are among the suppliers offering supercapacitor solutions and targeting data-center applications. While most of these companies have traditionally focused on ESS, renewable-energy, or other industrial applications, some have already begun entering the power-shelf market. We therefore believe Panasonic's upside from supercapacitors and CBUs may be relatively limited despite its BBU adjacency and established position with data-center customers. (*Exhibit 28, Exhibit 29*).
- **The transition to 800V DC could expand demand for full-rack power solutions, but Panasonic is unlikely to capture the opportunity alone.** Under an 800V DC architecture, BBU and CBU shelves can be consolidated into dedicated power racks supporting multiple high-density server racks and potentially enabling power delivery for 1MW-class rack configurations. However, competing suppliers are already developing integrated solutions for this architecture. Delta Electronics (covered by Alex Wang) has rolled out an 800V DC power-rack solution combining BBUs, CBUs, and power-supply units into a full-rack system serving multiple server racks. The transition to 800V DC therefore broadens the overall market opportunity, but the presence of established competitors with comparable system-integration capabilities limits the likelihood of disproportionate upside for Panasonic (*Exhibit 30*).

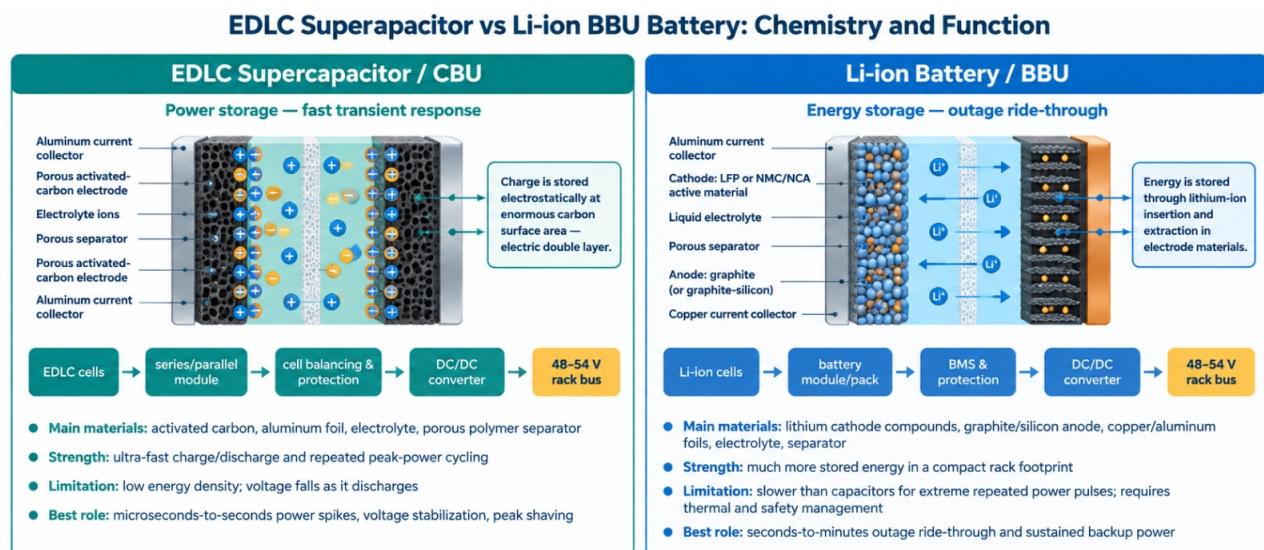
EXHIBIT 26: **Future execution issues: Panasonic was an early leader in Batteries, but was unable to hold their share against fast-growing competitors**



CATL, LGES, Samsung SDI (Covered by Neil Beveridge), BYD (Eunice Lee), SK Innovation is not covered by Bernstein
 Source: Bernstein Energy Storage Team

EXHIBIT 27: **Without Chinese players (CATL) entering, Panasonic's share was still challenged by Korean players**

CATL, LGES, Samsung SDI (Covered by Neil Beveridge), SK Innovation is not covered by Bernstein
 Source: Bernstein Energy Storage Team

EXHIBIT 28: **Supercapacitors or CBUs are using different materials that excel in peak shaving functions**

Source: Bernstein analysis

EXHIBIT 29: The supercapacitor industry is already crowded with component and module players established in providing DC peak shaving services

Company	Panasonic	Maxwell	VINATech	Nippon Chemi-Con	LS Material	TDK	Mitsubishi Electric	Musashi Energy Solutions
Domicile	Japan	USA	South Korea	Japan	South Korea	Japan	Japan	Japan
Business Description	Conglomerate with focus on Energy Solutions	Supercapacitor cell and module manufacturer	EDLC and hybrid supercapacitor cell and module manufacturer	EDLCs plus conductive-polymer, hybrid, and snap-in aluminum capacitors and modules manufacturer	Ultracapacitor cell and module manufacturer	Electronics component and module manufacturer	Conglomerate with focus on Data Center and Energy solutions	Hybrid Supercapacitor specialist, combining Lithium ion technology with EDLC construction.
AI DC Involvement	Developing CBU with supercapacitor functions for peak-shaving starting pilot testing in 2026	Developing new low-ESR ultracapacitor solutions for AIDC to deliver higher power with faster response	To reach KRW 1Tn (JPY115Bn/USD730Mn) in revenue by 2030E. Signed contract with Bloom Energy to supply supercap for AIDC through 2027	Announced new capacitor series in June 2026 with higher voltage, capacitance per volume and lifespan	Unveiled next generation capacitor for AI DC of 6Mn charging cycles (5 years) up from 1Mn	Expands ultra-capacitor series for AI servers and raise voltage rating to 500V for higher power density, reliability and minimal size	Accelerate deployment of next-gen 800V DC compatible power supply systems. Co-creation with Foxconn and Nvidia including supplying supercapacitors	Hybrid Supercapacitor is already solutions for peak shaving for AI DC within the racks
Competitive Edge	B2B adjacency with strong hyperscaler relationship and energy management reputation	Ultracapacitor specialist for automotive, renewable energy, industrial power and data centers	Established ultracapacitor specialist with existing industrial, transport and renewable expertise	Established ultracapacitor specialist with existing industrial, and renewable expertise	Producing rack type supercapacitor systems for Data Center using renewable energy since 2025	Expertise in electronics component including ultracapacitors, capacitors and other passive components used in data center	Leveraging extensive customer base and high reliability cultivated through already installed UPS systems and integrated engineering capabilities, and power electronics technologies	Specialised in proprietary material and processes to contain high energy density with rapid charge and discharge rates under restricted physical space.

Source: Company reports, Bernstein analysis

EXHIBIT 30: Delta's 800V DC solution is a dedicated BBU rack with consolidated BBU capacity for the relevant server racks



Source: Delta

WHAT DOES EROSION MEAN FINANCIALLY?

Competition affects Panasonic through two financial channels: share erosion reduces BBU revenue growth relative to the market, while selling-price pressure compresses margins. Our BOM analysis tests whether component-cost reductions can offset expected declines in BBU pricing as additional suppliers enter. In our base case, Panasonic's BBU adjusted operating margin declines from 20% in 2026 to 9% in 2032E. In our bear case, more aggressive price erosion causes margin to fall to 10% in 2029E, 3% in 2030E, and negative 3% in 2031E.

- **Our BOM framework estimates a margin-duration BBU module value of USD610.** Panasonic's BBU module combines battery cells, a battery-management system (BMS), power-conversion system, boards and electronics, thermal and mechanics. Battery cells represent the largest component, followed by the power-conversion system, thermal mechanics and then the BMS/control systems. **We estimate that one extended-duration module has a total modeled value of USD670, while a complete 5+1 BBU shelf has a modeled value of USD4,160 (Exhibit 31 - Exhibit 34).**
- **Battery cells account for 20-30% of the modeled module value, depending on the required support duration.** Our margin-duration case assumes 0.045kW of power per cell, 30 minutes of support, and a cell price of USD100/kWh. Under these assumptions, each cell has a modeled value of USD4.55. A module containing 66 cells, arranged as 11 cells in series and six in parallel, therefore has an estimated battery-cell value of USD150. Exhibit 30 also presents 60-minute and four-minute configurations to show the sensitivity to support duration. The battery-cell share of module value declines in higher-power BBUs, as power electronics account for a larger share.

- **The BMS accounts for 10-15% of the module value.** Based on the indicative semiconductor architecture and integration assumptions summarized in Exhibit 31, we estimate semiconductor inputs of around USD70 per module and a further USD10 of integration-related costs, resulting in an estimated BMS value of USD80.
- **The power-conversion system accounts for a further 25-35%.** We estimate the power conversion system would cost around USD170-200 per module as the largest category, this is largely from the MOSFET power semiconductor cost of ~USD60, capacitors and inductors taking up USD30 each. We assume another USD50 of additional integration and PCB costs.
- **Boards and Electronics account for the next 10-15%.** We estimate USD75 for this segment that includes PCB assembly and other electronic components.
- **Connectivity accounts for 5-10% of the module cost.** We estimate USD40 for this segment which comprises battery to board/ board to board connectors, protection packs, auxiliary sets and relay elements.
- **Thermal and Mechanics account for the remaining USD65 with 10-15%.** We estimate USD65 for the heat sinks, fans and rack mounts.
- **We estimate an extended-duration 5+1 BBU shelf value of USD4,160.** Each shelf contains six BBU modules, with five operating modules and one redundant module. Six modules at USD610 each produce a combined module value of USD3,660. Adding around USD500 for cooling fans, connectors, switches, the enclosure, and other shelf-level components results in an estimated total shelf value of USD4,160.
- **Additional qualified suppliers should increase pricing pressure across cells, BMS, and integrated modules.** Battery cells and BMS remain important sources of product differentiation, particularly in safety, power management, and system reliability. However, established battery manufacturers, module assemblers, and OEM/ODM suppliers possess relevant adjacent capabilities and can enter through internal development or supply-chain partnerships. As more suppliers qualify, we expect greater pricing pressure across battery cells, BMS-related functionality, and complete BBU modules, even though safety validation and system integration remain important barriers.
- We use the 30 minutes configuration as the main case for our BOM analysis below. We believe 30mins is the most likely configuration as DCs require some margin of safety from the 4mins that the OCP dictates. Our 60-minute configuration is the case for reconciling Panasonic's revenue with our market-size estimate. The BBU-4mins (120W) depicts Panasonic's future battery configuration of using 120W cell which requires a higher cost of 250kWh.
- **In our base case, BBU adjusted operating margin declines from 20% in 2026 to 9% in 2032E.** We model battery-cell pricing declining by 2% annually, in line with broader NCA and NCM pricing trends. BMS and other module prices remain stable through 2029E before declining by 5% annually from 2030E as supply constraints ease and additional suppliers enter. Operating costs decline by 2% annually from 2030E as the technology matures and incremental R&D and depreciation requirements stabilize. Under these assumptions, overall adjusted operating margin declines to 18% in 2028E and 2029E, 15% in 2030E, 12% in 2031E, and 9% in 2032E (*Exhibit 35*).
- **In our bear case, more aggressive price erosion causes BBU adjusted operating margin to turn negative by 2031E.** This scenario assumes stronger entry by Korean, Taiwanese, and Japanese competitors, allowing hyperscalers to exert greater pricing pressure. Battery-cell price declines accelerate by 5% YoY, this is consistent with the risk of commoditization seen in lithium-ion batteries, where pack and cell prices have declined at a 13% CAGR over the past decade. We also modeled the costs of BMS and other modules to grow at 2%YoY as the power semiconductor chips and other components cost could grow annually while Panasonic faces trouble to pass it on to CSPs in a competitive market environment. Under these assumptions, adjusted operating margin declines from 20% in 2026 to 10% in 2029E, 3% in 2030E, and negative 3% in 2031E (*Exhibit 36*).

EXHIBIT 31: **We estimate battery should take up 20-30% of total BBU costs**

BOM (USD)	BBU- Margin	BBU- Extended	BBU- 4Mins	BBU- 4Mins (120W)
Battery Cell				
Power per Cell (kW)	0.045	0.045	0.045	0.120
Duration (Mins)	30	60	4	4
Capacity per Cell (kWh)	0.02	0.05	0.00	0.01
Price per kWh (USD)	100	100	100	250
Battery Cell	2.27	4.55	0.30	2.00
Number of Cells in a Module	66	66	66	66
Battery Module	150	300	20	132
Battery Management System	80	80	80	80
Power Conversion System	200	200	200	200
Boards and Electronics	75	75	75	75
Connectivity	40	40	40	40
Thermal and Mechanics	65	65	65	65
Cost per Module	610	760	480	592
Number of Modules per Shelf	6	6	6	6
BBU Price per Shelf	3,660	4,560	2,880	3,552
Other Modules	500	500	500	500
Total BBU Shelf Cost	4,160	5,060	3,380	4,052
kW per shelf	18.0	18.0	18.0	47.5
Shelf cost per kWh	462.2	281.1	2816.7	1279.0

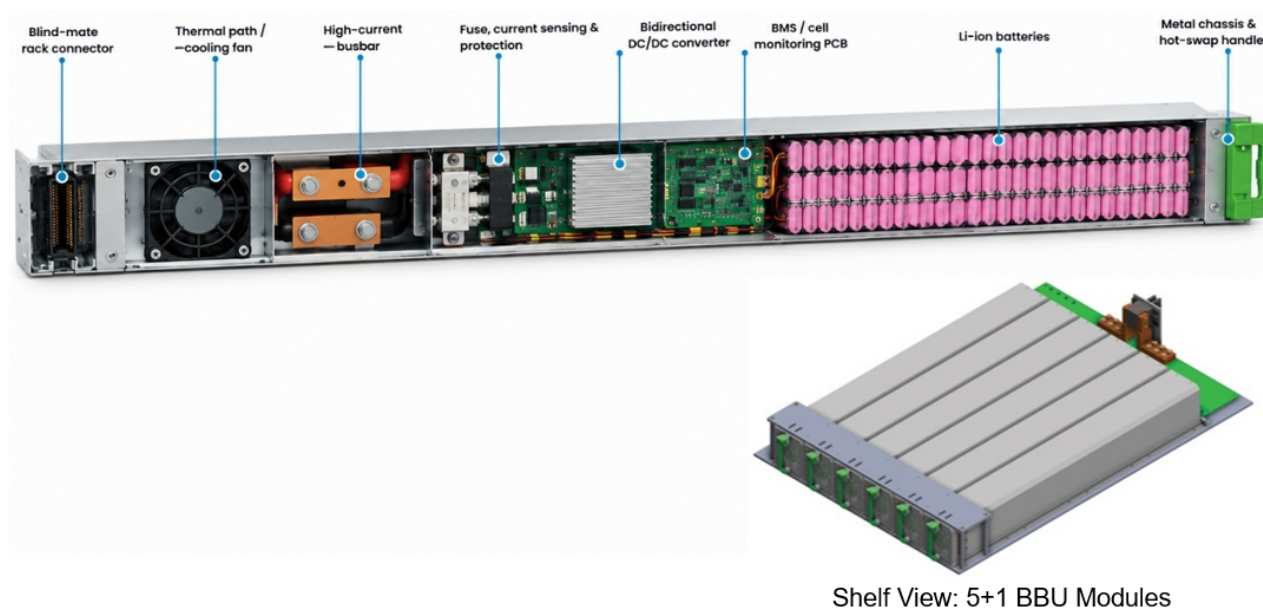
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 32: Our calculations show the BMS and Power Conversion systems take up the next largest cost profile after the battery cells

BBU Module	USD
Battery Management System	
LTC2971	9
LT8228	10
LT8551	4
ADBMS6948	15
MAX32625	6
MAX32690	20
ADA4522	15
MAX31760	1
24AA512T-1	2
ADM2561	7
Others	10
Discount (%)	15%
Total BMS Price	84.15
Power Conversion System	
MOSFET	60
Capacitors	30
Inductors and Magnetics	30
Others	50
Total Power Conversion System Price	170

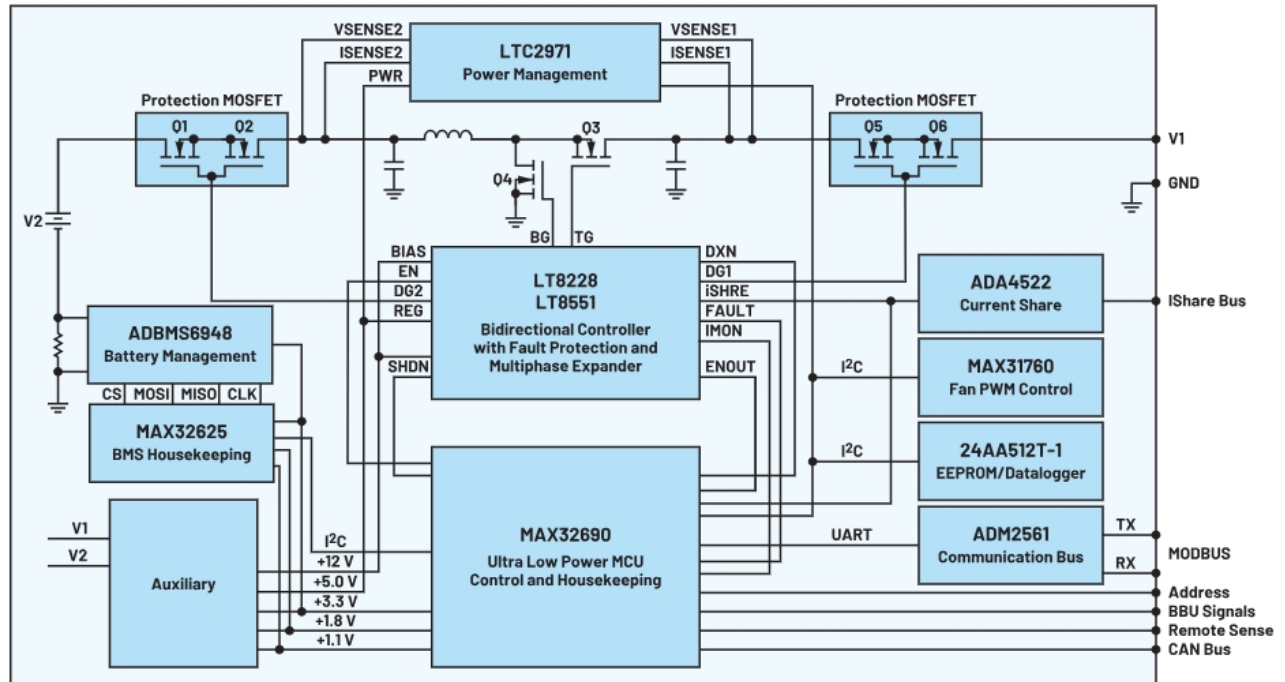
Component prices and model numbers are indicative estimates; Highlighted prices are estimates that include other items within the BMS like PCB, contactors, fuses etc. We take the lower end of pricing range as companies may negotiate from bulk purchases
Source: ADI, Rohm, Bernstein estimates and analysis

EXHIBIT 33: BBU modules mainly consist of battery cells, BMS, and power systems



Source: Open Compute Project, Bernstein analysis

EXHIBIT 34: Technology architecture within battery management systems in BBU module



Source: ADI, Bernstein analysis

EXHIBIT 35: Our simulated base case shows Panasonic's BBU OP Margin to fall from 20% to 9% in a scenario of moderate BBU price fall after 3 years from competition

BOM (USD)	2026	2027E	2028E	2029E	2030E	2031E	2032E
Battery Cell							
Capacity per Cell (kWh)	0.045	0.045	0.045	0.045	0.045	0.045	0.045
Duration (Mins)	30	30	30	30	30	30	30
Cost per kWh (USD)	100	98.0	96.0	94.1	92.2	90.4	88.6
Growth YoY (%)		-2%	-2%	-2%	-2%	-2%	-2%
Battery Cell	2.27	2.23	2.18	2.14	2.10	2.05	2.01
Number of Cells in a Pack	66	66	66	66	66	66	66
Battery Module	150	147	144	141	138	136	133
Battery Management System	80	80	80	80	78	77	75
Power Conversion System	200	200	200	200	196	192	188
Boards and Electronics	75	75	75	75	74	72	71
Connectivity	40	40	40	40	39	38	38
Thermal and Mechanics	65	65	65	65	64	62	61
Cost per Module	610	607	604	601	589	577	566
Number of Modules per Shelf	6	6	6	6	6	6	6
BBU Cost per Shelf	3,660	3,642	3,624	3,607	3,535	3,464	3,395
Other Modules	500	500	500	500	490	480	471
Total BBU Shelf Cost	4,160	4,142	4,124	4,107	4,025	3,944	3,866
kW per shelf	18.0	18.0	18.0	18.0	18.0	18.0	18.0
Shelf cost per kW	462.2	460.2	458.3	456.3	447.2	438.3	429.5
OP Margins (%)							
Battery Module	20%	16.3%	12.8%	10.5%	9.3%	8.2%	8.2%
Battery Management System	20%	20.0%	20.0%	20.0%	16.3%	12.8%	9.3%
Other Modules	20%	20.0%	20.0%	20.0%	16.3%	12.8%	9.3%
Overall Margin	20%	19%	18%	18%	15%	12%	9%
Price							
Battery Module	1,080	1,026	975	936	908	880	863
Battery Management System	576	576	576	576	547	520	494
Other Modules	3,336	3,336	3,336	3,336	3,169	3,011	2,860
Total Price	4,992	4,938	4,887	4,848	4,624	4,411	4,217
Cost Growth YoY (%)							
Battery Module		-2%	-2%	-2%	-2%	-2%	-2%
Battery Management System		0%	0%	0%	-2%	-2%	-2%
Other Modules		0%	0%	0%	-2%	-2%	-2%
Total Shelf		0%	0%	0%	-2%	-2%	-2%
Price Growth YoY (%)							
Battery Module		-5%	-5%	-4%	-3%	-3%	-2%
Battery Management System		0%	0%	0%	-5%	-5%	-5%
Other Modules		0%	0%	0%	-5%	-5%	-5%
Total Shelf		-1%	-1%	-1%	-5%	-5%	-4%

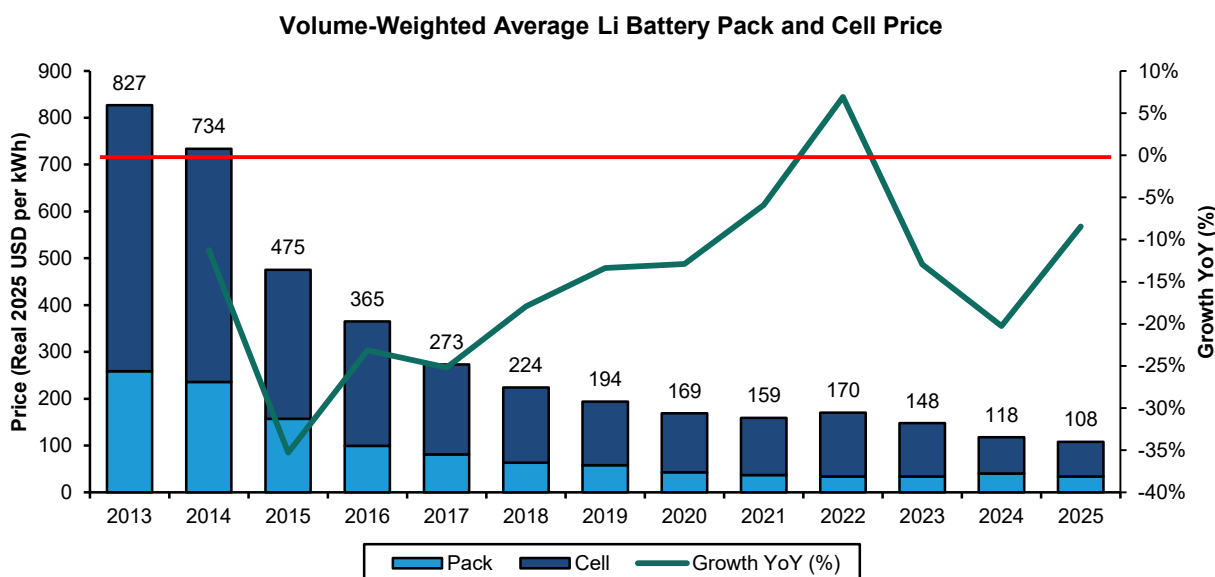
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 36: In the bear case, battery module prices can reduce by 5%YoY and costs for BMS and other modules could increase by 2% YoY

BOM (USD)	2026	2027E	2028E	2029E	2030E	2031E	2032E
Battery Cell							
Capacity per Cell (kWh)	0.045	0.045	0.045	0.045	0.045	0.045	0.045
Duration (Mins)	30	30	30	30	30	30	30
Cost per kWh (USD)	100	98.0	96.0	94.1	92.2	90.4	88.6
Growth YoY (%)		-2%	-2%	-2%	-2%	-2%	-2%
Battery Cell	2.27	2.23	2.18	2.14	2.10	2.05	2.01
Number of Cells in a Pack	66	66	66	66	66	66	66
Battery Module	150	147	144	141	138	136	133
Battery Management System	80	80	82	83	85	87	88
Power Conversion System	200	200	204	208	212	216	221
Boards and Electronics	75	75	77	78	80	81	83
Connectivity	40	40	41	42	42	43	44
Thermal and Mechanics	65	65	66	68	69	70	72
Cost per Module	610	607	613	620	627	634	641
Number of Modules per Shelf	6	6	6	6	6	6	6
BBU Cost per Shelf	3,660	3,642	3,680	3,719	3,759	3,801	3,845
Other Modules	500	500	510	520	531	541	552
Total BBU Shelf Cost	4,160	4,142	4,190	4,239	4,290	4,342	4,397
kW per shelf	18.0	18.0	18.0	18.0	18.0	18.0	18.0
Shelf cost per kW	462.2	460.2	465.5	471.0	476.6	482.5	488.5
OP Margins (%)							
Battery Module	20%	16.3%	12.8%	9.3%	6.0%	2.7%	-0.4%
Battery Management System	20%	20.0%	17.6%	9.6%	2.1%	-5.0%	-11.5%
Other Modules	20%	20.0%	17.6%	9.6%	2.1%	-5.0%	-11.5%
Overall Margin	20%	19%	17%	10%	3%	-4%	-9%
Price							
Battery Module	1,080	1,026	975	926	880	836	794
Battery Management System	576	576	576	547	520	494	469
Other Modules	3,336	3,336	3,336	3,169	3,011	2,860	2,717
Total Price	4,992	4,938	4,887	4,642	4,410	4,190	3,980
Cost Growth YoY (%)							
Battery Module		-2%	-2%	-2%	-2%	-2%	-2%
Battery Management System		0%	2%	2%	2%	2%	2%
Other Modules		0%	2%	2%	2%	2%	2%
Total Shelf		0%	1%	1%	1%	1%	1%
Price Growth YoY (%)							
Battery Module		-5%	-5%	-5%	-5%	-5%	-5%
Battery Management System		0%	0%	-5%	-5%	-5%	-5%
Other Modules		0%	0%	-5%	-5%	-5%	-5%
Total Shelf		-1%	-1%	-5%	-5%	-5%	-5%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 37: Lithium ion battery modules have fallen by ~13% CAGR over the last few years from capacity expansion and high competition



Source: BNEF, Bernstein analysis

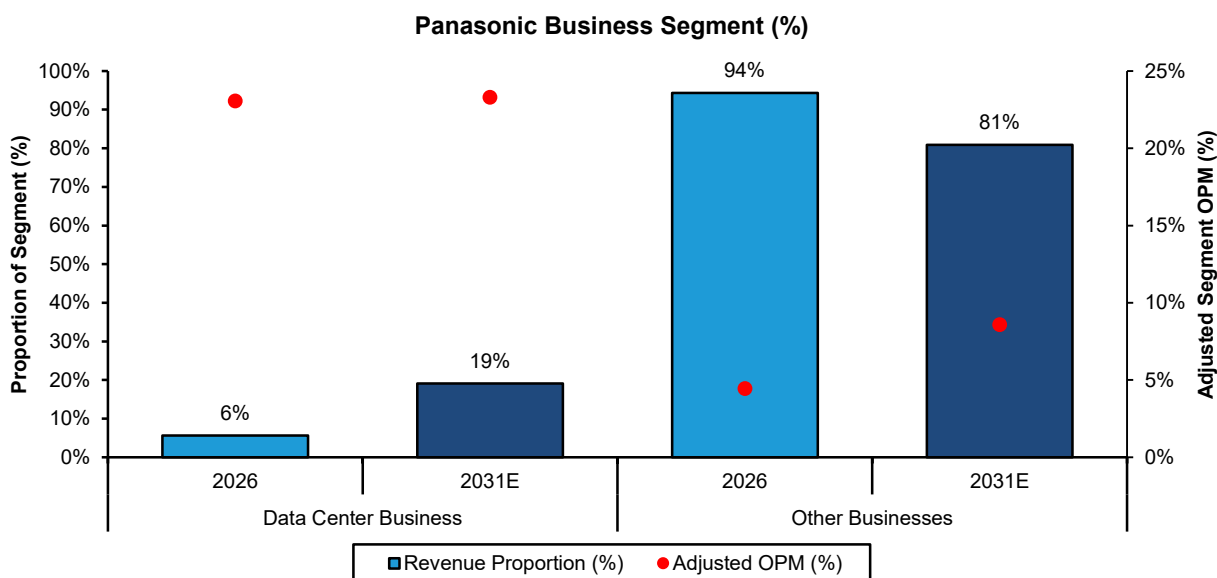
WHAT IS PRICED INTO THE STOCK?

Are we too pessimistic on valuation? We believe the current valuation already reflects a substantially stronger earnings outcome than our base case. Panasonic's data-center businesses generate adjusted operating margins above 20%, but account for only 6% of consolidated revenue, while its other businesses generate margins below 5%. Strong BBU growth must therefore offset slower growth and lower profitability across the rest of the portfolio. We estimate Panasonic's two-year blended-forward EBITDA growth at 14.5%, broadly in line with consensus, suggesting limited scope for earnings expectations alone to drive a further re-rating (*Exhibit 38*).

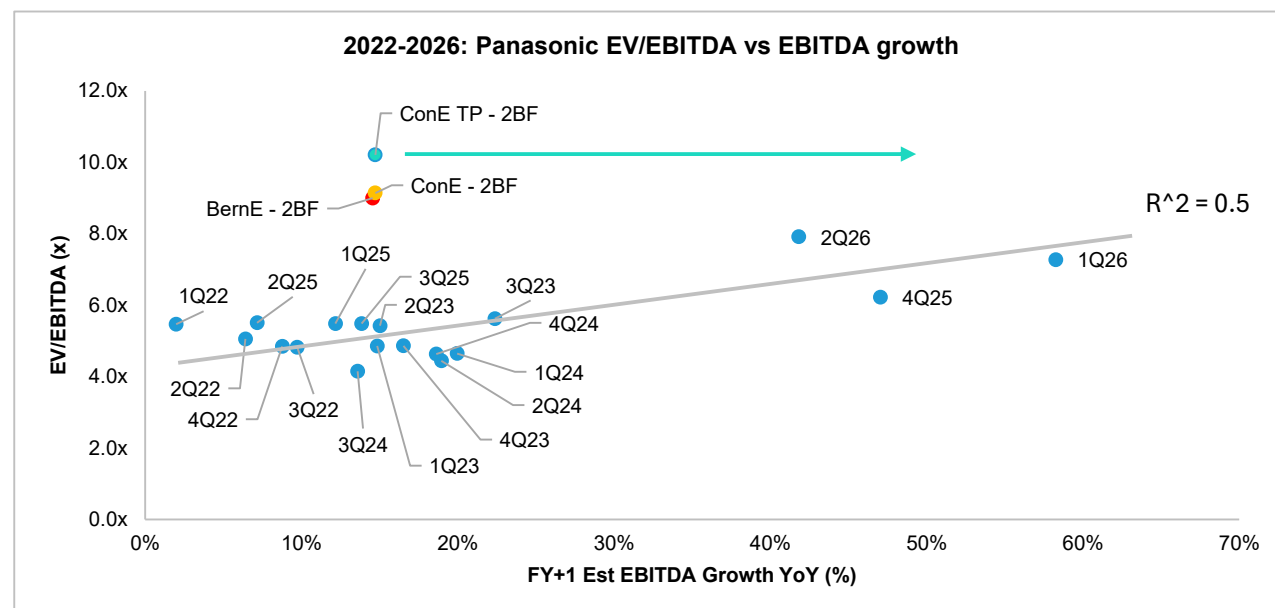
Panasonic's current valuation appears elevated relative to its expected earnings growth. The stock trades at 9x two-year blended-forward EV/EBITDA at the current share price and 10x at the average consensus target price of JPY5,100. Both points sit above Panasonic's historical relationship between expected EBITDA growth and EV/EBITDA, which shows a moderate correlation with an R^2 of 0.5. Based on this relationship, justifying a 9-10x multiple would require EBITDA growth above 35%, materially higher than our 14.5% estimate (*Exhibit 39*).

Achieving EBITDA growth above 35% would require a materially stronger operating outcome than our base case. Our sensitivity implies 250bp of incremental EBITDA-margin expansion in Energy, 100-170bp across the other businesses, 30% additional revenue growth in Industry and Energy, and 10% additional revenue growth across the remaining segments. These assumptions would require Panasonic to deliver stronger revenue growth and margin expansion across multiple businesses simultaneously, rather than relying on BBU growth alone (*Exhibit 40*).

Given our caution on future market share loss, we believe upside surprises are limited. The company has guided for JPY700bn BBU + CBU revenues by 2028E FY and almost JPY1tn by 2029E FY. From our conversation with investors, the consensus believe the market is sufficiently large for continuous upward revision in both the BBU and CBU which means expectations for guidance revision is quite high. Hence, even as the industry may have revisions, higher competition and high expectations may limit real upside surprises in the next twelve months.

EXHIBIT 38: **We expect Panasonic's revenue to grow by mid single digit in the next few years**

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 39: **We see Panasonic's EV/EBITDA multiple as being on the high side even as sell side EBITDA growth expectations remain high**

FY+1 Est EBITDA growth is the average of Bloomberg sell side consensus estimates for the 1 Year Forward EBITDA divided by trail 12m EBITDA. BernE-2BF represents our 2 year forward multiple on our blended 2 year forward EBITDA growth. ConE-2BF represents today's price divided by 2 year forward consensus EBITDA growth. ConE TP-2BF refers to average consensus TP of 5000 vs 2 year forward consensus EBITDA growth.

Source: Bloomberg, Bernstein estimates and analysis

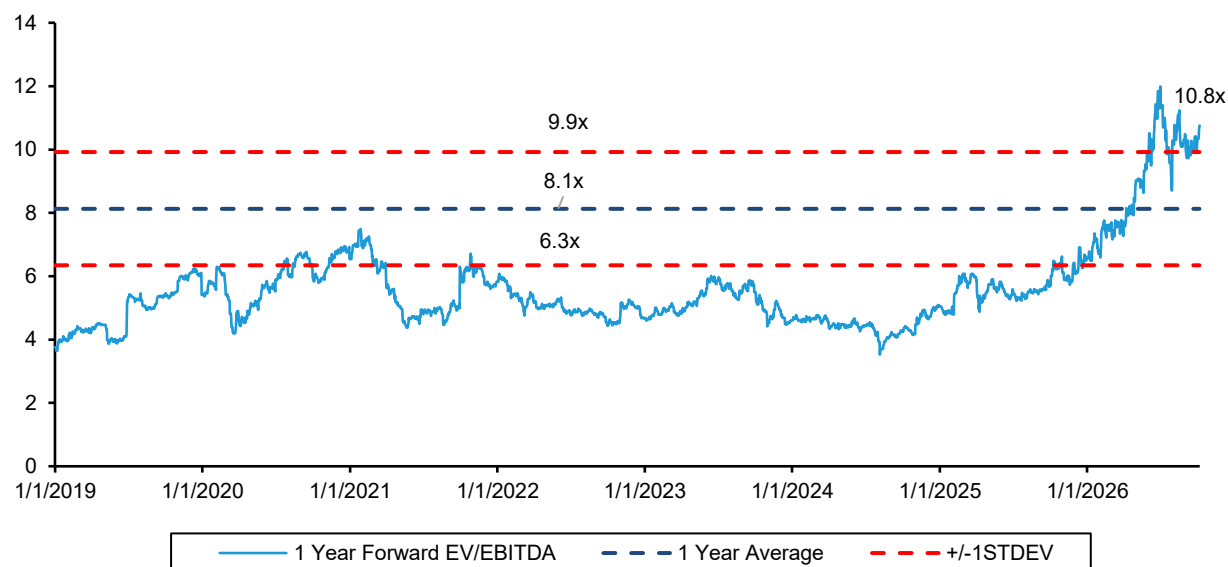
EXHIBIT 40: Hypothetically, to reach 35% EBITDA growth, one needs to believe 100-250bps incremental margin expansion and another 15% incremental revenue growth in FY28/29

	FY 26 EBITDA Margin (%)	BernE		Bull			
		2BF (FY28/29) EBITDA Margin (%)	FY+1 (FY28/29) EBITDA Growth YoY (%)	2BF (FY28/29) EBITDA Margin (%)	2BF (FY28/29) Incremental EBITDA Margin (%)	FY+1 (FY28/29) Incremental Revenue Growth YoY (%)	FY+1 (FY28/29) EBITDA Growth YoY (%)
Smart Life	4.0%	6.8%	4.3%	8.0%	1.3%	3.5%	28.3%
HVAC & CC	4.5%	6.3%	2.8%	8.0%	1.7%	6.5%	37.7%
Electric Works	9.6%	11.7%	6.4%	13.0%	1.3%	6.8%	26.4%
Connect	8.8%	11.4%	14.6%	13.0%	1.6%	3.6%	34.4%
Industry	11.4%	14.0%	2.0%	15.0%	1.0%	22.0%	31.7%
Energy	10.3%	17.5%	26.1%	20.0%	2.5%	11.1%	57.0%
Total	8.0%	14.0%	14.5%	15.2%	1.1%	15.2%	34.2%

We estimate the EBITDA margin by segment by using a 2-3 percentage point D&A assumption
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 41: Panasonic has seen significant multiple expansion in the last few years from ongoing transformation and AI sentiments

Panasonic 1-Year forward EV/EBITDA



Source: Bloomberg, Bernstein analysis

EXHIBIT 42: **We are slightly above consensus in the near term but inline with consensus further out**

Panasonic: Bernstein vs Consensus	2027E	2028E	2029E
JPY Billion			
Revenue			
Bernstein	8,154	8,737	9,240
Growth YoY	1.3%	7.2%	5.8%
Consensus	8,062	8,584	9,120
Growth YoY	0.2%	6.5%	6.2%
Bernstein vs. consensus	1.1%	1.7%	1.3%
Gross Profit			
Bernstein	2,608	2,795	2,991
Growth YoY	3.2%	7.2%	7.0%
Consensus	2,607	2,781	2,974
Growth YoY	3.2%	6.6%	6.9%
Bernstein vs. consensus	0.0%	0.5%	0.6%
Adjusted Operating Profit			
Bernstein	707	797	898
Growth YoY	58.1%	12.6%	12.7%
Consensus	709	816	947
Growth YoY	58.4%	15.2%	16.0%
Bernstein vs. consensus	-0.2%	-2.5%	-5.5%
EPS (JPY)			
Bernstein	204.60	251.87	284.92
Growth YoY	152.1%	23.1%	13.1%
Consensus	212.26	260.95	307.84
Growth YoY	161.5%	22.9%	18.0%
Bernstein vs. consensus	-3.7%	-3.6%	-8.0%

Source: Bloomberg, Company reports, Bernstein estimates and analysis

EXHIBIT 43: **Even a 25% increase in our 2031E earnings estimate, combined with an expansion in the blended multiple from 15x to 18x, would imply only a JPY1,000 per-share increase, representing 20% upside.**

Energy	Bear	Base	Bull
2031E Revenue	1,941	2,426	2,926
Adjusted OP Margin (%)		0%	
EBITDA Margin (%)	15%	19%	20%
EBITDA	291	461	585
EV/EBITDA Multiple (x)	12.0x	15.0x	18.0x
Enterprise Value	3,493	6,913	10,532
2028E Enterprise Value per Share (JPY)	1,065.58	2,108.97	3,213.08

Source: Company reports, Bernstein estimates and analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 44: Panasonic Income Statement

Income Statement

JPY billion	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027E	2028E	2029E	2030E	2031E
Revenue	7,344	7,982	8,003	7,491	6,699	7,389	8,379	8,496	8,458	8,049	8,154	8,737	9,240	9,743	10,278
Gross profit	2,187	2,339	2,266	2,151	1,975	2,082	2,261	2,494	2,629	2,527	2,608	2,795	2,991	3,213	3,444
Selling, general and administrative expenses	(1,843)	(1,938)	(1,939)	(1,864)	(1,668)	(1,725)	(1,947)	(2,104)	(2,161)	(2,079)	(1,901)	(1,999)	(2,093)	(2,171)	(2,305)
Adjusted Operating Profit	344	401	327	287	307	358	314	390	467	447	707	797	898	1,042	1,140
Other operating income (expenses)	(84)	(41)	63	(4)	(49)	(0)	(26)	(29)	(41)	(211)	(47)	18	18	18	18
Operating Profit	277	381	411	294	259	358	289	361	426	236	660	814	916	1,060	1,158
EBITDA	548	668	708	667	576	697	671	761	832	641	1,000	1,199	1,320	1,478	1,601
Net other financial income (expense)	(2)	(2)	5	(3)	2	3	28	64	60	27	0	(5)	(1)	5	12
Profit (loss) before income taxes	275	379	416	291	261	360	316	425	486	263	660	809	916	1,065	1,170
Income tax	(103)	(127)	(114)	(51)	(77)	(95)	(36)	40	(102)	(54)	(167)	(202)	(229)	(266)	(293)
Net income	172	252	303	240	184	265	281	465	384	209	493	607	687	799	878
EPS (JPY)	64.3	101.2	121.8	96.7	70.7	109.4	113.7	190.2	156.8	81.2	204.6	251.9	284.9	331.5	364.2

Margins

Gross profit	29.8%	29.3%	28.3%	28.7%	29.5%	28.2%	27.0%	29.4%	31.1%	31.4%	32.0%	32.0%	32.4%	33.0%	33.5%
Operating Profit	3.8%	4.8%	5.1%	3.9%	3.9%	4.8%	3.4%	4.2%	5.0%	2.9%	8.1%	9.3%	9.9%	10.9%	11.3%
Adjusted Operating Profit	4.7%	5.0%	4.1%	3.8%	4.6%	4.8%	3.7%	4.6%	5.5%	5.6%	8.7%	9.1%	9.7%	10.7%	11.1%
Profit before tax	3.7%	4.7%	5.2%	3.9%	3.9%	4.9%	3.8%	5.0%	5.7%	3.3%	8.1%	9.3%	9.9%	10.9%	11.4%
Net income	2.3%	3.2%	3.8%	3.2%	2.7%	3.6%	3.3%	5.5%	4.5%	2.6%	6.0%	6.9%	7.4%	8.2%	8.5%

Growth YoY

Revenue		8.7%	0.3%	-6.4%	-10.6%	10.3%	13.4%	1.4%	-0.5%	-4.8%	1.3%	7.2%	5.8%	5.4%	5.5%
Gross profit		7.0%	-3.1%	-5.1%	-8.2%	5.4%	8.6%	10.3%	5.4%	-3.9%	3.2%	7.2%	7.0%	7.4%	7.2%
Operating Profit		37.5%	8.1%	-28.6%	-12.0%	38.3%	-19.3%	25.1%	18.2%	-44.6%	179.2%	23.4%	12.5%	15.7%	9.2%
Adjusted Operating Profit		16.8%	-18.5%	-12.3%	7.1%	16.5%	-12.2%	24.2%	19.6%	-4.2%	58.1%	12.6%	12.7%	16.0%	9.4%
Profit before tax		37.6%	10.0%	-30.1%	-10.4%	38.2%	-12.2%	34.4%	14.4%	-45.9%	151.0%	22.6%	13.1%	16.3%	9.9%
Net income		46.2%	20.1%	-20.7%	-23.4%	44.3%	5.7%	65.9%	-17.4%	-45.6%	135.9%	23.1%	13.1%	16.3%	9.9%
EPS		57.3%	20.4%	-20.6%	-26.9%	54.7%	4.0%	67.2%	-17.5%	-48.2%	152.1%	23.1%	13.1%	16.3%	9.9%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 45: Panasonic Balance Sheet

Balance Sheet

JPY billion	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027E	2028E	2029E	2030E	2031E
Cash and cash equivalents	1,271	1,090	772	1,017	1,593	1,206	819	1,120	848	770	924	1,128	1,360	1,679	2,041
Trade receivables	847	1,039	1,191	1,051	1,194	1,325	1,323	1,361	1,316	1,380	1,432	1,499	1,598	1,684	1,775
Other financial assets	144	204	131	148	150	211	170	227	165	197	214	214	214	214	214
Inventories	806	989	1,016	794	833	1,133	1,289	1,209	1,022	1,066	1,235	1,293	1,368	1,432	1,476
Other current assets	137	165	163	426	153	157	202	236	264	467	520	520	520	520	520
Total current assets	3,205	3,486	3,274	3,436	3,923	4,031	3,803	4,153	3,615	3,880	4,324	4,655	5,060	5,529	6,026
Investments accounted for using the equity method	156	148	136	307	456	403	401	424	497	567	565	583	601	619	638
Other financial assets	162	166	216	215	231	213	243	207	186	213	220	220	220	220	220
Property, plant and equipment	1,323	1,374	1,324	1,035	1,062	1,115	1,172	1,559	1,902	2,244	2,646	2,952	3,232	3,513	3,798
Right-of-use assets	0	0	0	261	250	258	239	271	281	247	150	71	34	16	8
Other non-current assets	1,137	1,117	1,063	965	926	2,003	2,202	2,797	2,861	3,020	2,889	2,889	2,889	2,889	2,889
Total Non-current assets	2,778	2,805	2,740	2,783	2,924	3,992	4,257	5,258	5,728	6,293	6,470	6,715	6,976	7,259	7,554
Total Assets	5,983	6,291	6,014	6,219	6,847	8,024	8,060	9,411	9,343	10,172	10,795	11,370	12,036	12,788	13,579
Trade and other payables	956	1,146	1,151	970	1,046	1,664	1,663	1,690	1,506	1,497	1,614	1,690	1,788	1,871	1,928
Lease liabilities	0	0	8	64	59	61	60	69	62	59	58	58	58	58	58
Bonds and borrowings	177	375	390	251	310	433	159	262	107	186	139	139	139	139	139
Other financial liabilities	330	305	274	213	221	161	146	148	125	117	119	119	119	119	119
Other current liabilities	1,249	1,271	1,175	1,030	1,169	747	845	952	882	1,141	1,175	1,175	1,175	1,175	1,175
Total Current liabilities	2,712	3,098	2,989	2,616	2,804	3,065	2,873	3,122	2,683	2,999	3,106	3,182	3,280	3,363	3,421
Non-current liabilities	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bonds and borrowings	947	864	601	954	880	1,198	1,050	1,084	1,173	1,163	1,165	1,165	1,165	1,165	1,165
Lease liabilities	0	0	8	202	199	206	188	211	227	253	191	191	191	191	191
Other non-current liabilities	564	447	331	290	196	207	158	272	386	376	456	456	456	456	456
Total Non-current liabilities	1,511	1,311	940	1,447	1,274	1,611	1,396	1,568	1,786	1,791	1,812	1,812	1,812	1,812	1,812
Total Liabilities	4,223	4,409	3,929	4,063	4,079	4,676	4,270	4,689	4,468	4,790	4,918	4,994	5,092	5,175	5,233
Total equity attributable to owners of the parent	1,572	1,708	1,914	1,998	2,594	3,165	3,618	4,544	4,694	5,211	5,697	6,178	6,724	7,368	8,075
Non-controlling interests	188	175	171	158	174	182	172	178	180	171	180	199	220	245	272
Total Equity	1,760	1,882	2,085	2,156	2,769	3,347	3,790	4,722	4,875	5,382	5,877	6,376	6,944	7,613	8,347
Total Liabilities and Equity	5,983	6,291	6,014	6,219	6,847	8,024	8,060	9,411	9,343	10,172	10,795	11,370	12,036	12,788	13,579

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 46: **Panasonic Cash Flow Statement**

Cash Flow Statement

JPY billion	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027E	2028E	2029E	2030E	2031E
Net Income	172	252	303	240	184	265	281	465	384	209	493	607	687	799	878
Depreciation & amortization	271	288	296	373	318	339	382	400	406	404	340	385	404	418	443
Changes in working capital	19	(178)	(143)	8	(81)	(203)	(89)	181	(12)	(130)	(51)	(50)	(76)	(67)	(78)
Other	(77)	61	(252)	(191)	84	(149)	(53)	(180)	18	141	170	0	0	0	0
Cash flow from operations	385	423	204	430	504	253	521	867	796	624	951	942	1,015	1,150	1,243
Capex	(279)	(394)	(316)	(274)	(231)	(234)	(289)	(547)	(772)	(623)	(619)	(612)	(647)	(682)	(719)
Net proceeds from investment income	2	(9)	4	30	315	(6)	(2)	42	(15)	(23)	(17)	(18)	(18)	(18)	(18)
Other	(144)	(55)	119	37	93	(556)	(53)	(74)	(73)	39	(11)	0	0	0	0
Cash flow from investments	(420)	(459)	(193)	(206)	177	(796)	(344)	(579)	(860)	(607)	(647)	(629)	(665)	(700)	(738)
Cash flow from financing	295	(129)	(342)	48	(178)	59	(607)	(83)	(190)	(167)	(170)	(108)	(119)	(130)	(144)
Net effect of exchange rate changes	(2)	(17)	14	(28)	74	97	44	96	(18)	73	19	0	0	0	0
Net change in cash	258	(181)	(317)	244	577	(387)	(386)	300	(272)	(77)	153	205	231	319	361
Beginning cash	1,013	1,271	1,090	772	1,017	1,593	1,206	819	1,120	848	770	924	1,128	1,360	1,679
Ending cash	1,271	1,090	772	1,017	1,593	1,206	819	1,120	848	770	924	1,128	1,360	1,679	2,041

Source: Company reports, Bernstein estimates and analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Panasonic Holdings Corp**

We rate Panasonic Market-Perform, with target price of ¥4,500. We used an SOTP valuation and triangulate using an average of DCF (WACC: 7.5%, PGR: 3%) and 9.0x EV/EBITDA on 2 Year Blended Forward EBITDA.

RISKS**Panasonic Holdings Corp**

The downside risks include: 1) Faster than expected loss of BBU/CBU market share to new market entrants, 2) Continued loss of competitiveness in low-margin legacy businesses, including Smart Life and Electric Works. The upside risks include: 1) Technical barriers to entry in BBU/CBU is higher than expected and faster adoption of full-rack BBU and CBU solutions. 2) Successful restructuring of low-margin legacy businesses.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

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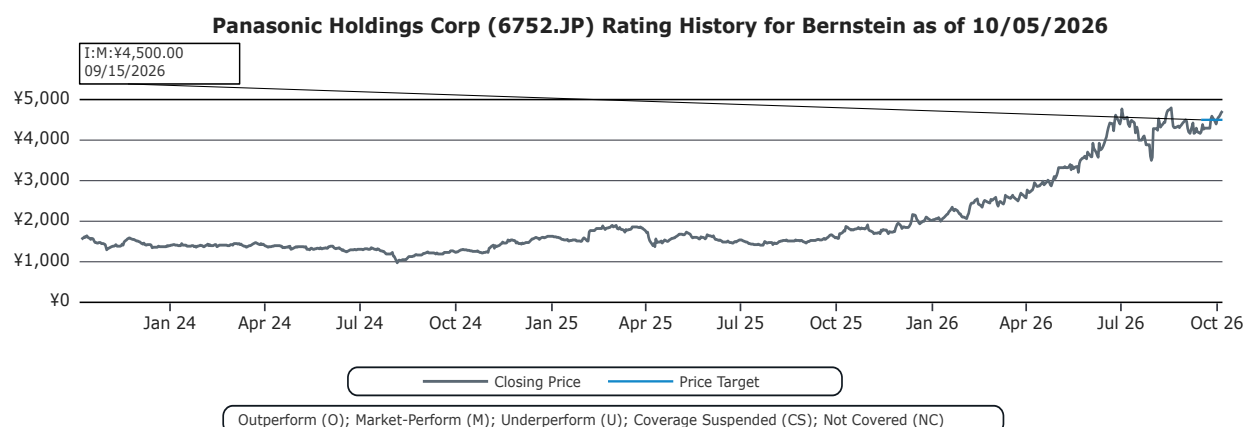
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